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Performance of CO_2 Cycle in the Context of Energy Transition

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Abstract

This thesis compares three power cycles for industrial waste heat recovery (WHR) in an Australian context — the Steam Rankine Cycle (SRC), the Supercritical CO_2 (sCO_2) Brayton cycle and the Transcritical CO_2 (tCO_2) Rankine cycle — under one consistent set of thermodynamic, sizing, cost and economic assumptions, applied to three representative sites: LNG gas processing in the Pilbara (200 °C, 30 kg/s), cement preheater exhaust in Gladstone (350 °C, 50 kg/s) and electric arc furnace off-gas in Sydney (500 °C, 20 kg/s). Four research questions address thermodynamic performance (RQ1), equipment sizing and capital cost (RQ2), techno-economic viability (RQ3) and off-design flexibility (RQ4). All models are implemented in Python with CoolProp and every result is regenerated by a single script.

Thirty findings are reported. With the recuperator energy balance limited by both streams — a second-law requirement that the interim analysis did not satisfy — the steam cycle has the highest thermal efficiency at every turbine inlet temperature from 150 to 600 °C (19.1–35.4 % against 13.3–30.1 % for tCO_2 and 5.3–30.1 % for sCO_2), and no crossover occurs within the WHR range. A 200-node integration of the tCO_2 heater against a waste-gas stream whose temperature is linear in heat duty shows a double pinch at both terminals with the driving force widest in the middle (42–65 K), so that two-point LMTD sizing overestimates heater area by 129–223 % when the pseudo-critical c_p peak lies inside the heater. The CO_2 cycles' decisive weakness in Australia is the heat sink: the tCO_2 baseline condenses at 14.3 °C, which a chiller could supply only by consuming 63–134 % of the net output, and the sCO_2 cycle loses 42 % of its 200 °C efficiency if its compressor inlet warms from 32 to 40 °C.

Integrated with each site's stream through a 10 K-approach heater, the steam boiler must run at 0.16–3.7 MPa rather than 10 MPa, and the tCO_2 cycle recovers the most power at 200 and 350 °C (723 and 3418 kW against 412 and 2639 kW for steam) because its cold heater inlet cools the gas furthest; the recuperated sCO_2 cycle recovers the least at every site. CO_2 turbines are 15–26 times more compact than the steam turbine, but at equal output steam is the cheapest plant at every site (AUD 2,600–4,400/kW at 1 MWe in 2026), tCO_2 costs 30–65 % more and sCO_2 two to three times more. On a fixed-source basis steam gives the lowest LCOE at all three sites (AUD 96, 35 and 31/MWh) and the highest NPV with 3–4-year discounted paybacks and 99.8–100 % Monte Carlo viability at 350 and 500 °C; tCO_2 is the

better investment at 200 °C (NPV AUD 1.71 M against 1.12 M) only where a natural cold sink exists; sCO_2 is not viable at any site. The tCO_2 cycle is the most flexible at part load (Marchionni score 0.78 against 0.73 for steam and 0.71 for sCO_2).

The thesis concludes that the figure of merit for WHR cycle selection is the net work recovered per kilogram of waste gas and its cost, not thermal efficiency, and that at Australian sink temperatures the steam Rankine cycle is the default choice above 300 °C, with transcritical CO_2 reserved for low-temperature sources where cold water is available.

Keywords: CO_2 power cycle; waste heat recovery; supercritical CO_2; transcritical CO_2; steam Rankine cycle; exergy analysis; pseudo-critical; pinch analysis; techno-economic analysis; LCOE; Australia; energy transition.

Executive Summary

This report evaluates the commercial viability of retrofitting waste heat recovery (WHR) power cycles at three Australian industrial sites: a Pilbara LNG train (200 °C), a Gladstone cement preheater (350 °C) and a Sydney electric arc furnace (500 °C). Three cycles — Steam Rankine (SRC), Supercritical CO_2 Brayton (sCO_2) and Transcritical CO_2 Rankine (tCO_2) — were modelled on one consistent basis, each sized to what it can recover from the site's exhaust stream and costed at that size in 2026 Australian dollars.

Key findings for capital allocation:

- **High- and medium-temperature WHR (350–500 °C):** The steam Rankine cycle is the clear choice. It is the most efficient of the three under Australian sink conditions, the cheapest to build (AUD 1,900–2,200/kW at the 2.3–2.6 MWe site scale), and the most bankable: LCOE of AUD 31–35/MWh against site prices of 90–100/MWh, discounted payback of 3.2–4.2 years, IRR of 28–36 %, and positive NPV in 99.8–100 % of 10,000 cost, price and utilisation scenarios.

- **Low-temperature WHR (200 °C):** Marginal for any cycle. Steam and tCO_2 both return about 12 % IRR with a one-in-three chance of loss, and depend on the Pilbara price premium. The tCO_2 cycle recovers 75 % more electricity from the stream and has 53 % more NPV, but only if a heat sink near 4 °C is available; supplying it with a chiller would consume more power than the plant makes.

- **Supercritical CO_2:** Not investable in its simple recuperated form at any of the three sites. With cooling water that cannot bring the compressor inlet below 36 °C, the cycle's back-work ratio is 31–64 %, its turbomachinery costs more than a whole steam plant, and its recuperated heater leaves most of the exhaust heat unrecovered. NPV is negative at two sites and marginal at the third.

- **Flexibility:** Where the heat source fluctuates (batch furnaces, intermittent trains), the tCO_2 cycle retains 56–67 % of its efficiency at one-fifth load against 38–66 % for steam, and starts up most simply.

Investment recommendation:
Target **steam Rankine** technology for Australian WHR sources at 350 °C and above. Treat 200 °C sources as site-specific studies, with transcritical CO_2 preferred

only where cold water or a co-located refrigeration load exists. Do not pursue simple supercritical CO_2 Brayton systems below 500 °C in Australian conditions. Evaluate every option on the electricity recovered from the actual stream and its cost, not on thermal efficiency.

Declaration by Author

The work presented in this Final Thesis is my own except where I have stated otherwise. The research questions, the choice of cycles, sites and modelling assumptions, the interpretation of all results, and the engineering conclusions and recommendations are mine, carried out under the supervision of Dr Alexander Klimenko at the University of Queensland.

In completing this thesis I used two generative artificial intelligence (AI) tools. Their use is disclosed in full here and in the AI Assistance log in Appendix C, in accordance with UQ's policy on the use of AI in assessment [36].

Tool 1 — Claude (Anthropic, 2026) [34]

I used Claude for the following purposes:

- Drafting initial paragraphs for the background and literature review (Chapters 1 and 2), which I checked against the primary sources and revised.
- LaTeX typesetting of the document, equations, tables and figure environments, and the Matplotlib styling of the figures.
- Reviewing the Python models I wrote for the Interim Report. This review identified three errors in my code: a recuperator energy balance that violated the second law, a heater gas-temperature profile that was linear in node index rather than in heat duty, and heater temperature-difference expressions in the cost model that produced unphysical values. It also identified that my turbine cost coefficients could not be traced to the source I had cited.
- Restructuring my model scripts into the `co2whr` package described in Appendix A, implementing the corrections above, adding the fixed-source site integration, the consistent cost basis of Table 3.2, discounted payback and the breakeven analysis. I ran the resulting code, checked the energy and exergy balances, traced every correlation to its source, and compared the outputs with my original results and with the literature.
- Drafting the text of Chapters 4–9, the abstract and the executive summary from the model outputs, which I then reviewed, corrected where necessary and approved. The findings, their numbering and their interpretation were settled by me.

The cycle models of the Interim Report (steam, sCO_2 and tCO_2 state-point mod-

els, exergy balances, pinch analysis, SP/VFR sizing, LCOE/NPV and Monte Carlo, Stodola part-load model) were written by me and remain the physical basis of the corrected package. Every number in this thesis was produced by running that package, and I can reproduce and explain every result and every line of the code independently of any AI tool.

Tool 2 — GitHub Copilot (Microsoft, 2026) [35]

I used GitHub Copilot for autocomplete assistance with Python syntax while writing the original model files — specifically CoolProp `PropsSI` argument strings and NumPy and Matplotlib syntax. GitHub Copilot did not generate any model logic, result or text appearing in this thesis.

I have read and understood the University of Queensland's Student Code of Conduct and the UQ policy on academic integrity and the use of AI in assessment [36]. I confirm that this submission complies with those policies and that I am able to demonstrate my learning, skills and abilities regarding all content in this submission independently of any AI tool.

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The open-source thermophysical property library CoolProp [18], the Python scientific computing ecosystem (NumPy, SciPy, Matplotlib) and NIST REFPROP data [19] were essential to the modelling work.

Two generative AI tools were used in preparing this thesis, in accordance with UQ's policy on AI in assessment [36] and the UQ Library guidance on acknowledging and referencing AI [37]. Claude (Anthropic, 2026) [34] assisted with drafting, typesetting, code review and the restructuring of the model code, as set out in the Declaration and in Appendix C; GitHub Copilot (Microsoft, 2026) [35] assisted with Python syntax. The research design, the interpretation of results and the conclusions are my own.

List of Abbreviations

Abbreviation	Full Term
A2EP	Australian Alliance for Energy Productivity
AEMO	Australian Energy Market Operator
ARENA	Australian Renewable Energy Agency
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BWR	Back-Work Ratio
CEPCI	Chemical Engineering Plant Cost Index
CRF	Capital Recovery Factor
EAF	Electric Arc Furnace
HEOS	Helmholtz Energy Equation of State
HX	Heat Exchanger
IRR	Internal Rate of Return
LCOE	Levelised Cost of Electricity
LNG	Liquefied Natural Gas
MAUD, kAUD	Million / thousand Australian dollars (2026)
LMTD	Log-Mean Temperature Difference
NETL	National Energy Technology Laboratory (USA)
NIST	National Institute of Standards and Technology
NPV	Net Present Value
NTU	Number of Transfer Units
PCHE	Printed Circuit Heat Exchanger
PPTD	Pinch Point Temperature Difference
REFPROP	Reference Fluid Thermodynamic and Transport Properties
RQ	Research Question
sCO_2	Supercritical Carbon Dioxide
SP	Size Parameter (turbomachinery)
SRC	Steam Rankine Cycle
tCO_2	Transcritical Carbon Dioxide
T-H	Temperature–Enthalpy (diagram)
TEC	Total Erected Cost
TIT	Turbine Inlet Temperature

Abbreviation	Full Term
VFR	Volumetric Flow Ratio
WHR	Waste Heat Recovery

List of Symbols

Symbol	Definition	Units
a, b	Power-law cost correlation coefficients	—
c_p	Specific heat at constant pressure	kJ/(kg·K)
$\dot{E}_x$	Exergy rate	kW
$\dot{E}_{x,dest}$	Exergy destruction rate	kW
$\dot{m}$	Mass flow rate	kg/s
$\dot{Q}$	Heat transfer rate	kW
$\dot{S}_{gen}$	Entropy generation rate	kW/K
$\dot{W}$	Power (work rate)	kW
ε	Recuperator effectiveness	—
η_{ex}	Exergy efficiency	—
η_{is}	Isentropic efficiency	—
η_{th}	Thermal efficiency	—
f	Part-load mass-flow fraction	—
f_T	Heat exchanger temperature correction factor	—
h	Specific enthalpy	kJ/kg
h_0	Dead-state specific enthalpy	kJ/kg
P	Pressure	MPa
P_c	Critical pressure of CO_2	MPa
P_0	Dead-state pressure	kPa
s	Specific entropy	kJ/(kg·K)
s_0	Dead-state specific entropy	kJ/(kg·K)
T	Temperature	°C or K
T_0	Dead-state temperature	K
T_c	Critical temperature of CO_2	°C
T_{pc}	Pseudo-critical temperature	°C
UA	Overall heat transfer conductance	W/K
U	Overall heat transfer coefficient	W/(m^2·K)
v	Specific volume	m^3/kg
w_{net}	Net specific work output	kJ/kg
w_{gas}	Net work per kilogram of waste gas	kJ/kg

Symbol	Definition	Units
$c_{p,gas}$	Specific heat of the waste gas (1.10)	kJ/(kg·K)
T_{src}	Waste-gas source temperature	°C
$T_{gas,out}$	Waste-gas outlet temperature	°C
ex	Specific flow exergy	kJ/kg
C_{cap}	Capital cost	AUD
C_{OM}	Annual O&M cost	AUD/yr
E_{gen}	Annual electricity generation	MWh/yr
CRF	Capital recovery factor	—
SP	Turbine size parameter	$\mathrm{m}\cdot(\mathrm{kJ/kg})^{-0.25}$
VFR	Volumetric flow ratio	—

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1. Introduction

1.1 Background and Context

Somewhere between 20 and 50 % of the fuel burned in industrial facilities never does useful work. It departs the process as hot exhaust gas, cooling water, or radiated heat and is rejected to the atmosphere [1, 2]. Across manufacturing and processing in aggregate — cement kilns operating above 900 °C, electric arc steel furnaces, gas compression trains, refinery distillation columns — these losses are large enough that converting even a modest fraction into electricity would materially reduce both operating costs and carbon dioxide emissions without requiring new primary energy infrastructure [3].

Australia has particular reason to take industrial waste heat seriously. Cement manufacture, liquefied natural gas (LNG) processing, and electric arc furnace steelmaking rank among the country's most energy-intensive industries, all operating at temperatures at which power cycle generation is thermodynamically viable. The Australian Renewable Energy Agency's industrial transformation programme identified waste heat recovery as one of the most cost-competitive near-term decarbonisation options available to heavy industry [4]. The complication is that the majority of published engineering analysis of WHR power cycles was conducted in the United States or Europe, with economic boundary conditions — electricity prices, ambient temperatures, labour rates, and equipment supply chains — that do not translate directly to Australian conditions. This creates a gap, and the present thesis is designed to partially close it.

Three power cycle technologies compete for this application. The *Steam Rankine Cycle* (SRC) is the mature baseline. Subcritical SRC plants typically convert 33–38 % of heat input to electricity, with supercritical steam pushing that toward 45–48 % at high turbine inlet temperatures [6]. Below approximately 300 °C, efficiency declines steeply as the saturation pressure falls [7], which is the thermodynamic opening for CO_2-based cycles.

Both CO_2 variants exploit the low critical point of carbon dioxide: 31.1 °C and 7.38 MPa. The *Supercritical CO_2 (sCO_2) Brayton cycle* maintains CO_2 above both values throughout the entire loop, eliminating phase change entirely and enabling compact turbomachinery. The cost is that compressing a gas always requires more

work than pumping a liquid; at lower source temperatures, the compressor consumes a prohibitive share of turbine output. The *Transcritical CO_2 (tCO_2) Rankine cycle* handles this by condensing CO_2 to a liquid on the low-pressure side and pumping it above the critical pressure, reducing back-work ratio significantly. The operational constraint is that condensation requires a heat sink below 31.1 °C, which is not always achievable in warmer Australian locations [11, 12].

This thesis compares all three cycles simultaneously under consistent Australian-specific assumptions, a combination not previously addressed in the published literature. The outcome is a practical cycle selection framework for Australian industrial operators considering WHR investment.

1.2 Definitions

The following terms are defined here for unambiguous use throughout.

Rankine Cycle (RC):

A closed thermodynamic cycle in which the working fluid is pumped as a liquid (1→2), heated and evaporated at constant pressure (2→3), expanded through a turbine (3→4), and condensed back to liquid (4→1). The SRC uses water operating subcritically [5].

Supercritical CO_2 (sCO_2) Brayton Cycle:

A closed power cycle in which CO_2 remains above its critical pressure (7.38 MPa) and temperature (31.1 °C) throughout. No phase change occurs. Near-critical CO_2 is substantially less compressible than an ideal gas, reducing compressor work relative to a conventional gas turbine, but back-work ratio remains higher than for liquid-pump cycles [8, 9].

Transcritical CO_2 (tCO_2) Rankine Cycle:

A cycle that crosses the CO_2 saturation boundary. CO_2 condenses to a subcritical liquid on the low-pressure side, is pumped above critical pressure, heated supercritically, and expanded. Liquid-phase compression keeps pump work low (back-work ratio 12–23 % across the modelled range). The requirement for a heat sink below 31.1 °C is the main operational constraint [11, 12].

Pseudo-Critical Temperature (T_{pc}):

The temperature at which the isobaric specific heat c_p reaches its local maximum at a given supercritical pressure. CO_2 undergoes no phase transition above P_c, but c_p peaks sharply in a narrow band near the saturation line extension, creating anomalously high heat capacity. At 12.8 MPa, $T_{pc} \approx 57$ °C with

peak $c_p = 4.4\,\text{kJ/(kg·K)}$; at 20 MPa, $T_{pc} \approx 76\,^\circ\text{C}$ with peak $c_p = 2.6\,\text{kJ/(kg·K)}$. This non-uniform thermophysical behaviour is the central transitional effect addressed in Sections 4.4 and 5.3.

Waste Heat Recovery (WHR):

Capturing thermal energy from industrial exhaust or process cooling streams that would otherwise be rejected to the environment, and converting it to electricity or useful work [2].

Thermal Efficiency (η_{th}):

Net work output divided by total heat input: $\eta_{th} = \dot{W}_{net}/\dot{Q}_{in}$. A first-law metric that quantifies energy conversion but reveals nothing about the thermodynamic quality of the losses.

Exergy Efficiency (η_{ex}):

Net work output divided by the exergy of the heat input: $\eta_{ex} = \dot{W}_{net}/\dot{E}_{x,in}$. A second-law metric that accounts for the thermodynamic quality of the energy supplied, not merely its magnitude.

Back-Work Ratio (BWR):

The fraction of gross turbine output consumed by the compression device: $\text{BWR} = \dot{W}_{pump}/\dot{W}_{turbine}$. For SRC: below 1.5 %. For sCO_2 Brayton: 27–80 % across the modelled range. For tCO_2 Rankine: 12–23 % (higher than the literature-reported 2–5 % owing to the $P_{low} = 5$ MPa baseline; sensitivity analysed in Section 4.6).

Pinch Point Temperature Difference (PPTD):

The minimum local temperature difference between hot and cold streams at any cross-section within a heat exchanger. PPTD governs the trade-off between heat recovery and exchanger surface area [5].

Levelised Cost of Electricity (LCOE):

A lifecycle-averaged cost in AUD/MWh representing the electricity price at which a project breaks even over its operational life, after accounting for capital cost, annual operating expenditure, capacity factor, and the time value of money [26].

Specific Flow Exergy (ex):

The maximum work extractable from a fluid stream brought reversibly to the dead state ($T_0 = 25\,^\circ\text{C}$, $P_0 = 101.325\,\text{kPa}$): $ex = (h - h_0) - T_0(s - s_0)$.

Size Parameter (SP):

A turbomachinery compactness metric: $\text{SP} = \dot{V}_{out}^{0.5}/\Delta h_{is}^{0.25}$. A smaller SP indi-

cates a more compact turbine for a given net power output [20].

Stodola Ellipse Law:

An empirical approximation describing how turbine mass flow and pressure ratio evolve under off-design operation. Here applied in the simplified form $\eta_{od} = \eta_0 \cdot [1 - 0.3(1 - f)^2]$ where f is the mass-flow fraction [17].

1.3 Knowledge Gaps

Five specific gaps in the published literature motivated the design of this study:

Gap 1 — No three-cycle benchmark under consistent assumptions. Existing studies compare at most two technologies simultaneously, under different boundary conditions and cost assumptions, making cross-study inference unreliable [11, 13, 14].

Gap 2 — Combined exergy and pinch analysis is absent for tCO_2. The application of combined exergy-pinch methodology to the tCO_2 Rankine cycle in a WHR context has not been reported in the accessible literature [15, 16].

Gap 3 — Australian industrial conditions are unrepresented in the techno-economic literature. All published LCOE and NPV analyses for CO_2 WHR cycles use US or European economic parameters. Australian electricity prices, ambient temperatures, and supply chain structures differ sufficiently to make direct transplanting of overseas conclusions unreliable [2, 4].

Gap 4 — Transitional thermophysical effects in tCO_2 heat exchangers are not modelled. The sharp c_p peak near the pseudo-critical temperature creates a strongly non-uniform temperature-difference profile along the heater. Two-state-point energy balances and two-point LMTD sizing cannot represent this profile, so the location of the pinch and the required heat exchanger area are misestimated. This gap was specifically identified by the project supervisor.

Gap 5 — Off-design data for tCO_2 cycles are sparse. Characterisation of part-load performance and operational flexibility for tCO_2 Rankine cycles remains limited, with Yang et al. [27] representing one of few recent contributions.

1.4 Research Questions and Aim

To compare the Supercritical CO_2 Brayton cycle and Transcritical CO_2 Rankine cycle against the Steam Rankine Cycle for industrial waste heat recovery, using thermodynamic modelling and techno-economic analysis, and to determine under what conditions CO_2 cycles represent the better choice in terms of efficiency, capital cost, and economic return within the Australian energy transition context.

RQ1 — Thermodynamic Performance:
How do thermal efficiency, exergy efficiency, and back-work ratio compare for SRC, sCO_2 Brayton, and tCO_2 Rankine across WHR source temperatures of 150–600 °C, including sensitivity to recuperator effectiveness and condensing pressure?

RQ2 — System Compactness and Capital Cost:
How compact are the three cycles, and what does this imply for capital cost (AUD/kW) across the 100 kW to 5 MW range at the three Australian case study conditions?

RQ3 — Australian Techno-Economic Analysis:
For three representative Australian industrial sites, what are the LCOE, NPV, payback period, and probability of economic viability for each cycle under realistic uncertainty?

RQ4 — Off-Design and Part-Load Performance:
When the heat source fluctuates, how does each cycle's efficiency degrade, and what is the relative operational flexibility of the three technologies?

1.5 Scope and Limitations

The thermodynamic models are steady-state, single-fluid, zero-pressure-drop representations of idealised cycles. Real-fluid properties are evaluated using CoolProp 6.x [18] with the HEOS backend, which provides high accuracy near the CO_2 critical point. The WHR heat source is modelled as a dry ideal gas with constant specific heat; real flue gas composition effects are not considered. All three cycles use fixed isentropic efficiencies ($\eta_{turbine} = 0.85$, $\eta_{pump/compressor} = 0.80$), a known simplification explicitly quantified in Section 8.5. Capital cost correlations from Weiland et al. [23] and Turton et al. [21] carry inherent uncertainty of order ±30 % and are applied below their validated size range; this is explicitly propagated through Monte

Carlo analysis. Off-design modelling uses the simplified Stodola parabolic efficiency correction rather than full turbomachinery characteristic maps.

1.6 Thesis Structure

Chapter 2 reviews the literature on WHR power cycles, transitional CO_2 thermophysics, and Australian industrial context. Chapter 3 describes the methodology for all four research questions. Chapters 4–7 report results for RQ1–RQ4 respectively. Chapter 8 synthesises across research questions, situates the findings within the broader literature, and explicitly addresses limitations. Chapter 9 states the answers to each research question and identifies priority directions for future work.

2. Literature Review

2.1 Industrial Waste Heat Recovery

The quantity of recoverable industrial waste heat is large. Papapetrou et al. [2] estimated the technically available resource across European manufacturing at 300–900 TWh per year, with roughly one-third arising above 300 °C. Johnson et al. [1] placed US industrial waste heat losses at 20–50 % of primary energy input, consistent with the European estimate, with high-temperature sources arising primarily from metallurgy and cement, medium-grade from refinery and glass, and low-grade from compressed gas systems and cooling circuits.

For Australia, the Australian Renewable Energy Agency's (ARENA) industrial transformation study identified WHR as one of the most cost-competitive near-term options for decarbonising energy-intensive industry [4]. The Australian Alliance for Energy Productivity (A2EP) [30] compiled an industry-specific database confirming WHR opportunities at Australian cement, LNG, and steel facilities at temperatures closely matching the three case studies defined for this thesis. A distinguishing feature of the Australian context is the co-existence of high industrial electricity prices (particularly in the isolated Pilbara grid at AUD 130/MWh or above [31]) with climatic conditions that constrain condenser temperatures — a combination that creates simultaneously attractive economics and operational challenges for low-temperature cycle condensation.

2.2 Steam Rankine Cycle

The SRC has driven industrial power generation since the nineteenth century and its design practice is codified across a wide capacity range. Tian et al. [6] surveyed SRC across WHR temperatures, finding thermal efficiencies of 33–38 % for subcritical configurations and approaching 48 % for supercritical steam above 550 °C. The well-documented limitation below 300 °C [7] — where falling saturation pressure constrains maximum turbine inlet conditions — is the primary motivation for CO_2 cycle investigation in the medium-grade WHR range.

From a second-law perspective, Moran and Shapiro [5] identify the steam boiler as the dominant SRC loss mechanism. Water boils at a fixed saturation temperature while the heat source is at a higher temperature, so a sustained temperature

difference is maintained throughout the two-phase region, generating entropy continuously. CO_2 cycles address this by heating supercritically and non-isothermally, tracking the source temperature profile more closely. This thermodynamic advantage is the foundation of the CO_2 case, though it must be weighed against differences in component cost and operational behaviour.

Capital cost is the SRC's competitive strength. Turton et al. [21] document established shell-and-tube boiler cost correlations that benefit significantly from economies of scale. Baglietto et al. [14] confirmed that at larger WHR scales, steam's mature supply chain erodes the sCO_2 capital cost advantage, a finding corroborated by the present study's RQ2 analysis.

2.3 Supercritical CO_2 Brayton Cycle

Dostal's 2004 MIT dissertation [8] established the foundational analysis of near-critical compression and demonstrated the compactness advantage over steam through detailed cycle modelling. Crespi et al. [9] reviewed major sCO_2 configurations — simple recuperated, recompression, split-flow — and documented thermal efficiencies of 40–50 % at turbine inlet temperatures above 500 °C. Ahn et al. [10] assessed technology readiness, identifying turbomachinery as the critical engineering challenge and noting the absence of demonstrated commercial-scale plant.

The WHR case for sCO_2 is best illustrated by Kizilkan [13], who found sCO_2 producing 9,363 kW net versus 8,275 kW for SRC at a 350 °C cement WHR source — a 13.1 % improvement at respective thermal efficiencies of 27.6 % and 24.2 %. Baglietto et al. [14] added the critical qualifier that the sCO_2 capital cost advantage over steam erodes at larger scales due to the immaturity of the PCHE supply chain.

Below approximately 350 °C, sCO_2 compressor back-work rises to 20–40 % and continues upward as temperature falls [9]. This is the primary thermodynamic motivation for the tCO_2 cycle.

2.4 Transcritical CO_2 Rankine Cycle

Replacing gas compression with liquid pumping reduces BWR to the 2–5 % range reported in the literature at moderate-to-high temperatures, while retaining the supercritical heat addition advantage over SRC. Jiang et al. [11] reported up to 4.98 % better thermal efficiency for a recompression tCO_2 configuration over equivalent sCO_2 Brayton, which informed the working hypothesis of this study. Lee and Kim [12] provided experimental validation of a small-scale tCO_2 Rankine system,

supporting the modelling assumptions of comparable simulation work.

Yang et al. [27] examined off-design behaviour of tCO_2 cycles for low-grade WHR, quantifying efficiency degradation at part-load that had not previously been characterised — directly relevant to RQ4 of this study. Liu et al. [28] demonstrated that optimal high-side pressure increases with turbine inlet temperature, a result incorporated into the parametric sweep methodology described in Chapter 3. Wang et al. [29] extended this to variable heat source conditions.

A practical constraint specific to Australian conditions deserves emphasis. The tCO_2 cycle requires condensation below 31.1 °C. In the Pilbara in summer (ambient 35–42 °C [33]) or coastal Queensland, this requires supplementary mechanical or evaporative cooling, adding capital cost and parasitic load not accounted for in European or North American studies. This study treats this constraint explicitly in the RQ3 economic analysis.

2.5 Transitional Thermophysical Effects in CO_2 Heat Exchangers

The thermodynamic consequences of the pseudo-critical c_p peak in heat exchanger design have attracted increasing attention since Dostal's original analysis [8]. Sarkar [15] identified the primary heat exchanger as the dominant exergy destruction site in the sCO_2 recompression cycle, attributing this to the strongly non-uniform temperature difference between source and working fluid along the exchanger length. Nami et al. [16] demonstrated that combined exergy-economic optimisation surfaces improvement opportunities invisible to first-law analysis.

For the tCO_2 cycle specifically, CO_2 enters the supercritical heater at pump exit temperature (24–32 °C at 12–20 MPa), passes through the pseudo-critical region with its pronounced c_p peak, and emerges at turbine inlet temperature. Because c_p is largest near T_{pc}, a large fraction of the total heat transfer is absorbed within a narrow temperature band: the CO_2 temperature rises slowly through the pseudo-critical region and then quickly beyond it. Against a waste-gas stream whose temperature falls linearly with heat duty, this produces a temperature-difference profile that is pinched at the terminals and widest in the middle of the exchanger — the opposite of the uniform driving force that a two-point LMTD calculation assumes. The quantitative consequence for heat exchanger sizing, addressed in Section 5.3, had not been reported in the literature accessed for this study, representing Gap 4 identified in Section 1.3.

2.6 Capital Cost Correlations for WHR Power Systems

The published cost literature for small-scale CO_2 power systems is limited and predominantly US-based. The most complete public set of correlations is that of Weiland, Lance and Pidaparti [23], derived from US DOE project vendor quotes (2017 USD). It gives the PCHE/recuperator cost as $C_{HX} = 49.45 \cdot UA^{0.7544} \cdot f_T$, where UA is in W/K and f_T is a temperature penalty factor that exceeds unity above 550 °C; the radial sCO_2 turbine cost as $406{,}200 \cdot \dot{W}_{sh}^{0.8}$ ($\dot{W}_{sh}$ in MW); and direct air coolers as $32.88 \cdot UA^{0.75}$. Weiland et al. report uncertainty bands of −31 % to +38 % on the recuperator correlation and −32 % to +51 % on the radial turbine correlation, and the turbine data span 8–35 MW, so application at the sub-5 MW scale of industrial WHR is an extrapolation. Weiland and Thimsen [22] set out the steady-state modelling assumptions and constraints (compressor inlet conditions, recuperator limits) that underpin such cycle models. No comparable public correlation exists for small steam turbine-generators; the US DOE industrial benchmark [24] of roughly USD 900/kW at 150 kW falling to USD 200/kW at 2 MW (equipment only) is the most widely cited small-scale reference.

Turton et al. [21] provide cost functions for conventional equipment (centrifugal pumps and compressors, shell-and-tube heat exchangers) in 2001 USD (CEPCI 397), which can be escalated via the Chemical Engineering Plant Cost Index (CEPCI) [25]. The 2017 annual CEPCI of 567.5 and the estimated 2026 value of approximately 830 imply an escalation factor of 1.46× for the Weiland correlations. Conversion of USD to AUD at the April 2026 spot rate (1 USD = 1.395 AUD, RBA [32]) completes the localisation.

2.7 Off-Design Analysis Methods

The Stodola ellipse law provides a widely-used approximation for turbine off-design mass flow as a function of inlet/outlet pressures [17]. In its simplified form applicable to efficiency estimation, it approximates part-load isentropic efficiency degradation as a parabolic function of mass-flow fraction, consistent with the incidence-loss behaviour of radial turbines. Marchionni et al. [17] reviewed off-design performance of advanced thermodynamic cycles and proposed a multi-attribute flexibility scoring framework that weights turndown ratio, part-load efficiency retention, and startup complexity — the framework adopted for RQ4 in this study.

Yang et al. [27] examined off-design performance of tCO_2 cycles specifically and found efficiency degradation at part-load had not previously been quantified. Their

results inform the design-point calibration of the Stodola correction used in Chapter 7.

2.8 Summary and Gap Identification

The literature reviewed here is extensive but incomplete in five specific respects identified in Section 1.3. The thermodynamic promise of CO_2 WHR cycles is well established. What is missing is a single, unified comparison of all three cycle architectures under consistent Australian- specific assumptions, incorporating transitional thermophysical effects, full capital cost estimation, and off-design analysis. The methodology described in Chapter 3 addresses all five gaps within a single undergraduate thesis scope.

3. Methodology

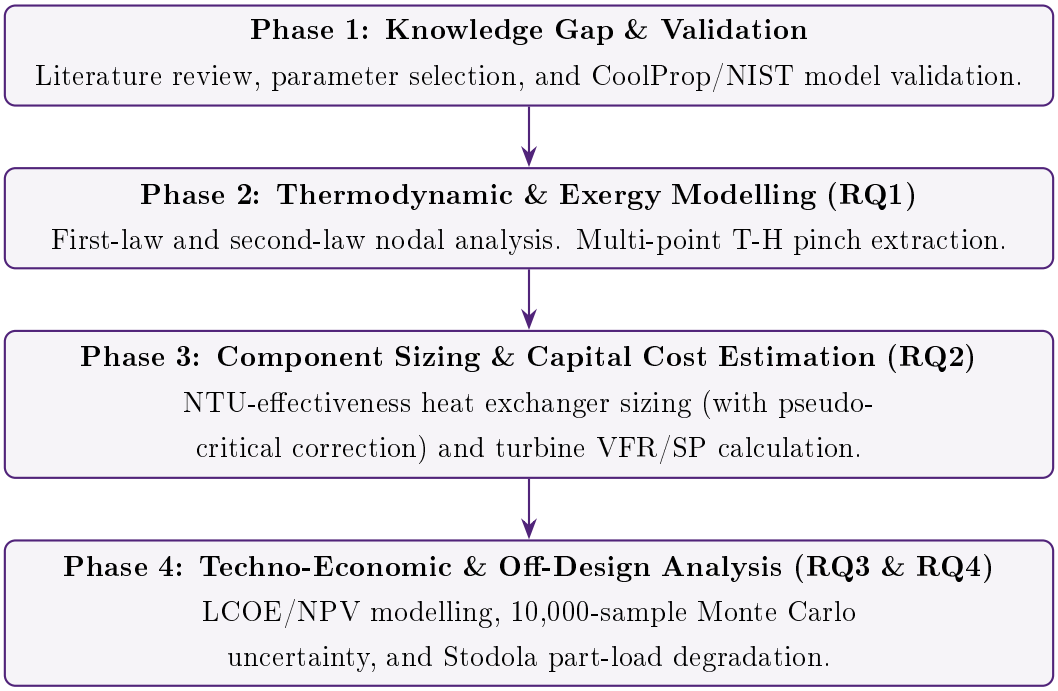


Figure 3.1: Overview of the interconnected four-phase methodology used to address RQ1–RQ4.

The research was structured in four phases corresponding to the four research questions. The approach was set out in the thesis proposal, substantially expanded in the Interim Report, and revised during the final modelling work described here. Three revisions made after the Interim Report are material to the results and are flagged where they arise: (i) the recuperator energy balance is limited by both streams (Section 3.2.2), which lowers the CO_2 cycle efficiencies reported at the interim stage; (ii) every primary heater is integrated with the waste-gas stream using a duty-linear gas profile and a 10 K approach (Section 3.5); and (iii) the techno-economic comparison is made on a fixed-source basis (Section 3.7).

3.1 Phase 1 — Literature Review

A structured search was conducted across Scopus, Web of Science, and Google Scholar following the PRISMA framework, covering WHR publications from 2005 to 2025 with five foundational pre-2005 sources included on merit (Dostal 2004; Short et al. 1995; Moran & Shapiro 2018; Turton et al. 2012; Bell et al. 2014). Coverage

was organised around six themes: cycle thermodynamics and variants; experimental and simulation performance data; transitional and pseudo-critical CO_2 thermophysics; heat exchanger technology for high-pressure CO_2; capital and operating cost data; and Australian industrial waste heat availability. Thirty-two references were retained after relevance screening; 29 are cited in the body of the thesis.

3.2 Thermodynamic Modelling (RQ1)

3.2.1 Cycle Architecture and State-Point Models

Steady-state energy balance models for all three cycles were implemented in Python 3.9 using the CoolProp 6.x library [18] for thermodynamic property evaluation. The HEOS (Helmholtz Energy Equation of State) backend was used throughout in preference to the default BICUBIC interpolation because of its superior accuracy near the CO_2 critical point, where the tCO_2 pump exit and the sCO_2 compressor inlet both operate. At each state point four properties are extracted: specific enthalpy (h), specific entropy (s), density (ρ), and temperature (T).

Energy balances at each state point follow standard open-system forms. For turbomachinery:

$$\dot{W} = \dot{m}(h_{in} - h_{out}) \tag{3.1}$$

For heat exchangers:

$$\dot{Q} = \dot{m}(h_{out} - h_{in}) \tag{3.2}$$

Isentropic efficiency is applied as:

$$h_{out} = h_{in} + \frac{h_{out,is} - h_{in}}{\eta_{is}} \quad \text{(compression)} \tag{3.3}$$

$$h_{out} = h_{in} - \eta_{is}(h_{in} - h_{out,is}) \quad \text{(expansion)} \tag{3.4}$$

Values of $\eta_{turbine} = 0.85$ and $\eta_{pump/compressor} = 0.80$ are applied uniformly to all three cycles, ensuring genuine comparability. These values are standard for preliminary analysis of small-to-medium scale turbomachinery [20] and represent a known simplification addressed further in Section 8.5.

All heat exchangers and pipe runs are modelled as isobaric processes ($\Delta P = 0$) in the thermodynamic analysis phase. Real frictional pressure drop — particularly significant in CO_2 PCHEs — is an identified limitation acknowledged and quantified in Section 8.5.

3.2.2 Cycle-Specific Parametric Assumptions

The tCO_2 high-side pressure is scaled linearly from 12 MPa at 150 °C to 20 MPa at 350 °C and above, following the optimal pressure relationship of Liu et al. [28] ($P_{high} = \min[20, \max(12,\ 8 + 0.048\,(T - 100))]$ MPa). Low-side pressure is held at 5 MPa for the baseline, corresponding to a condenser temperature of 14.3 °C. The recuperator effectiveness baseline is $\varepsilon = 0.70$ for tCO_2 and $\varepsilon = 0.85$ for sCO_2.

The recuperator effectiveness is applied to the stream with the smaller enthalpy capacity, so that neither stream can be driven past the inlet temperature of the other:

$$Q_{recup} = \varepsilon \cdot \min\left[\, h_{cold}(T_{hot,in}) - h_{cold,in},\ h_{hot,in} - h_{hot}(T_{cold,in}) \,\right] \tag{3.5}$$

This is the standard ε-NTU definition [5]. It matters for CO_2 because c_p differs strongly between the high- and low-pressure sides near the critical point: at the CS2 condition the cold (12.8 MPa) side can absorb 251 kJ/kg before reaching the turbine exhaust temperature, whereas the hot (5 MPa) side can release only 113 kJ/kg before reaching the pump exit temperature. Defining ε on the cold side alone, as was done in the Interim Report, transfers 176 kJ/kg and drives the hot stream to 14 °C — 10 K *below* the stream it is heating — which violates the second law. Equation (3.5) is used throughout this thesis, and the resulting efficiencies are lower than the interim values (Section 4.2). The recuperator is also discretised in 200 enthalpy steps to confirm that no internal temperature crossover occurs (Section 3.6).

The sCO_2 cycle uses $P_{high} = 20$ MPa and $P_{low} = 8$ MPa, with the compressor inlet set to $T_c + 5$ K (35.98 °C) to guarantee operation in the supercritical region and maintain the near-critical compressibility benefit.

In the parametric sweep the SRC uses $P_{high} = \min[10\,\text{MPa},\ 0.95 \times P_{sat}(T_{TIT})]$, i.e. the highest subcritical pressure that leaves the turbine inlet slightly superheated, and $P_{low} = 10$ kPa (condenser temperature 45.8 °C), a typical industrial condensing pressure. This sweep characterises each cycle as a function of turbine inlet temperature only; when the cycles are integrated with a real waste-gas stream (Section 3.6) the steam pressure is instead chosen by the boiler pinch constraint, which gives substantially lower pressures.

3.2.3 Model Validation

CoolProp outputs were cross-validated against NIST Webbook data [19] at two SRC state points (Table 3.1). Both deviations lie well within the 1 % acceptance criterion. CoolProp evaluates the CO_2 critical point as $T_c = 30.98$ °C and $P_c = 7.377$ MPa,

matching NIST REFPROP to within 0.04 %.

Table 3.1: SRC model validation against NIST Webbook reference data.

State point	NIST h (kJ/kg)	Model h (kJ/kg)	NIST s (kJ/kg·K)	Deviation
Saturated liquid, 10 kPa	191.83	191.81	0.6493	0.010 %
Superheated steam, 400 °C, 10 MPa	3096.5	3097.45	6.2120	0.031 %

The global exergy balance (Eq. 3.11) closes to machine precision ($< 10^{-12}$ kJ/kg) for all three cycles, confirming internal consistency of the second-law model.

3.2.4 Parametric Sweep

The primary sweep covers TIT from 150 to 600 °C in 25 °C steps. At each temperature, η_{th}, net specific work, BWR, and exergy destruction by component are recorded. Separate sweeps for recuperator effectiveness (Section 3.4) and low-side pressure (Section 3.4) extend the analysis beyond the single-parameter baseline.

3.3 Exergy Analysis (RQ1)

An exergy balance is applied to every cycle component at a common turbine inlet temperature of 400 °C and at the three site design points (Section 3.6). The dead state is $T_0 = 298.15$ K, $P_0 = 101.325$ kPa (ASHRAE standard, confirmed with supervisor). Because the three cycles circulate very different mass flows per unit of output, component destruction is reported both per kilogram of working fluid and per kilojoule of net work; only the latter is a like-for-like basis.

Specific flow exergy:

$$ex = (h - h_0) - T_0(s - s_0) \tag{3.6}$$

For adiabatic turbomachinery:

$$\dot{E}_{x,dest} = T_0\,\dot{m}(s_{out} - s_{in}) \tag{3.7}$$

For heat exchangers receiving heat from source at T_{source}:

$$ex_{heat,in} = Q_{in}\left(1 - \frac{T_0}{T_{source}}\right) \tag{3.8}$$

$$\dot{E}_{x,dest,HX} = ex_{heat,in} - (ex_{out} - ex_{in}) \tag{3.9}$$

For condensers rejecting heat to environment:

$$\dot{E}_{x,dest,cond} = ex_{in} - ex_{out} \tag{3.10}$$

Global balance closure check:

$$\dot{W}_{net} + \dot{E}_{x,dest,total} = ex_{heat,in} \tag{3.11}$$

3.4 Sensitivity Analysis (RQ1)

3.4.1 Recuperator Effectiveness Sweep

The tCO_2 cycle is evaluated at four recuperator effectiveness values ($\varepsilon = 0, 0.50, 0.70, 0.85$) at each of the three site design points, with all other parameters held at baseline. For each ε value the heater integration of Section 3.5 is repeated to determine how recuperation raises the heater inlet temperature and therefore the lowest temperature to which the waste gas can be cooled. The quantity of interest for waste heat recovery is not efficiency alone but the net work obtained per kilogram of waste gas, $w_{gas} = \eta_{th}\, c_{p,gas}\, (T_{src} - T_{gas,out})$, which combines both effects.

3.4.2 Low-Side Pressure Sweep

The tCO_2 cycle is evaluated at $P_{low} = 3.5, 4.0, 5.0, 6.0$ and $7.0\,\text{MPa}$ with $\varepsilon = 0.70$ fixed, at each site design point. Condensing temperature, η_{th}, BWR, achievable gas outlet temperature and w_{gas} are recorded. The feasibility of each P_{low} value is assessed against the cooling-water temperatures available at the three Australian sites, and the parasitic load of mechanical chilling, where it would be required, is estimated from the condenser duty and a chiller coefficient of performance of 4–6.

3.5 Transitional Effects: Multi-Point T-H Heater Integration (RQ1/RQ2)

Following supervisor feedback to model transitional thermophysical effects, the two-state-point energy balance of the primary heater is replaced by a nodal integration

along the exchanger. The working fluid is discretised in 200 equal enthalpy steps between the heater inlet and the turbine inlet, and the temperature and c_p are evaluated from real-fluid properties at every node, which resolves the pseudo-critical peak. The same procedure is applied to the steam boiler (where CoolProp returns the saturation temperature through the two-phase region) and to the sCO_2 heater, so that all three cycles are integrated with the waste-gas stream by one method.

The waste gas is treated as an ideal gas with constant c_p in counter-flow. Its temperature therefore falls *linearly with heat duty*, not with position or with node index:

$$T_{gas}(q) = T_{gas,out} + (T_{src} - T_{gas,out})\,\frac{q}{Q} \tag{3.12}$$

where q is the cumulative heat transferred from the cold end and Q the total heater duty per kilogram of working fluid. The distinction matters: a profile that is linear in node index implicitly assumes the gas gives up equal heat per node, which is false wherever the working fluid's c_p varies. The Interim Report used an index-linear profile, which produced a spurious constant 10 K difference along the whole heater; the corrected profile is used here.

Two constraints close the problem. The turbine inlet is set 10 K below the source temperature (hot-end approach), and the gas outlet temperature is found by bisection so that the minimum local temperature difference equals the 10 K approach:

$$\mathrm{PPTD} = \min_i [T_{gas,i} - T_{wf,i}] = 10\,\mathrm{K} \tag{3.13}$$

This is the maximum heat that the stream can give up to that cycle. Because the heater inlet temperature of the working fluid is the floor for the gas outlet, a cycle whose heater inlet is hot (a recuperated sCO_2 cycle, for example) can recover less of the stream than one whose heater inlet is cold. The conductance is then accumulated segment by segment,

$$UA = \sum_i \frac{\Delta Q_i}{\mathrm{LMTD}_i}, \qquad \mathrm{LMTD}_i = \frac{\Delta T_i - \Delta T_{i+1}}{\ln(\Delta T_i / \Delta T_{i+1})} \tag{3.14}$$

which is exact for counter-flow with piecewise-linear profiles. The two-point LMTD result, $UA_{2pt} = Q/\mathrm{LMTD}(\Delta T_{cold}, \Delta T_{hot})$, is computed alongside for comparison (Section 5.3).

3.6 Equipment Sizing and Capital Cost (RQ2)

3.6.1 Site Design Points

Each Australian site (Table 3.3) is a fixed waste-gas stream with source temperature T_{src} and mass flow $\dot{m}_{gas}$. Each cycle is integrated with the stream through the heater model of Section 3.5. For the CO_2 cycles the pressures follow the RQ1 rules. For the steam cycle the boiler pressure is a free design variable: the evaporation temperature fixes an internal pinch at the start of boiling, so a higher pressure raises cycle efficiency but prevents the gas from being cooled as far. The pressure is therefore chosen by a golden-section search to maximise the net work per kilogram of waste gas,

$$w_{gas} = \eta_{th}\, c_{p,gas}\, (T_{src} - T_{gas,out}), \qquad \dot{W}_{site} = \dot{m}_{gas}\, w_{gas} \tag{3.15}$$

with $c_{p,gas} = 1.10\,\mathrm{kJ/(kg{\cdot}K)}$. Equation (3.15) is the quantity a plant owner actually receives from a given stream; it rewards both conversion efficiency and depth of heat recovery, and it is used throughout RQ2–RQ4. The recuperator of each CO_2 cycle is discretised in the same way as the heater (200 enthalpy steps with real properties on both sides) to obtain its UA and to check that no internal temperature crossover occurs. The heat-rejection exchanger (condenser or gas cooler) is assigned a mean temperature difference of 10 K to the cooling medium for all three cycles.

3.6.2 Turbomachinery Compactness

Turbine size parameter and volumetric flow ratio are calculated from Macchi and Astolfi [20]:

$$\mathrm{SP} = \frac{\dot{V}_{out}^{0.5}}{\Delta h_{is}^{0.25}}, \qquad \mathrm{VFR} = \frac{v_{out}}{v_{in}} \tag{3.16}$$

where $\dot{V}_{out}$ is the volumetric flow rate at turbine exhaust ($\mathrm{m^3/s}$ per kg/s of working fluid) and Δh_{is} is the isentropic enthalpy drop (kJ/kg). A VFR below approximately 5 is taken as the threshold for single-stage radial turbine feasibility [20]. Both are evaluated at the site design points.

3.6.3 Heat Exchanger Area

For the illustrative comparison of sizing methods, the tCO_2 heater area is reported as $A = UA/U$ with the overall coefficient computed from the PCHE resistance network

$$\frac{1}{U} = \frac{1}{h_{CO_2}} + \frac{t_{wall}}{k_{wall}} + \frac{1}{h_{gas}} \tag{3.17}$$

with $h_{CO_2} = 2000\,\mathrm{W/m^2K}$ (supercritical CO_2 side, conservative PCHE estimate from Dostal [8]), $h_{gas} = 500\,\mathrm{W/m^2K}$ (exhaust gas forced convection), $t_{wall} = 2\,\mathrm{mm}$, and $k_{wall} = 15\,\mathrm{W/mK}$ (SS316), giving $U = 380\,\mathrm{W/m^2K}$. The comparison is made at $\varepsilon = 0$ (pump exit to turbine inlet), which places the pseudo-critical c_p peak inside the heater and maximises the divergence between the segmented and two-point results. Capital costs are computed from UA directly, so the value of U does not affect the economics.

3.6.4 Capital Cost Estimation

Every cycle is costed with the same method and the same correlations, applied component by component to the site design point scaled to the required net output (Table 3.2). CO_2 turbines and all heat exchangers use the NETL correlations of Weiland, Lance and Pidaparti [23]; the steam turbine uses a power law fitted to the US DOE small-turbine benchmarks [24]; pumps and the sCO_2 compressor use Turton et al. [21]. Turbine cost is evaluated on *shaft* power ($\dot{W}_{net}/(1 - \mathrm{BWR})$), which penalises the high back-work ratio of the sCO_2 cycle correctly. The condenser or cooler, omitted at the interim stage, is included for all three cycles because low-efficiency WHR plants reject three to six units of heat per unit of electricity.

Table 3.2: Capital cost basis. All correlations escalated to 2026 by CEPCI (2001: 397; 2012: 584.6; 2017: 567.5; 2026: ≈830 [25]) and converted at 1 USD = 1.395 AUD [32]; +35 % installation on all equipment [21].

Component	Correlation (cost year)	Scaling variable	Validity / note
Primary heater, recuperator	$49.45\,UA^{0.7544}\,f_T$ (2017) [23]	UA (W/K)	1.6×10^5–2.1×10^8 W/K; $f_T=1$ below 550 °C
Condenser / cooler	$32.88\,UA^{0.75}$ (2017) [23]	UA (W/K)	10 K mean ΔT assumed
CO_2 turbine	$406{,}200\,\dot{W}_{sh}^{0.8}$ (2017) [23]	shaft MW	8–35 MW; extrapolated below
Steam turbine	$a\,\dot{W}_{sh}^{b}$, fitted to USD 900/kW at 150 kW and USD 200/kW at 2 MW (2012) [24]	shaft kW	back-pressure benchmark; condensing units cost more
sCO_2 compressor	Turton centrifugal compressor (2001) [21]	kW	450–3000 kW
Feed / CO_2 pump	Turton centrifugal pump (2001) [21]	kW	1–300 kW; no pressure factor
Generator	$108{,}900\,\dot{W}_e^{0.5463}$ (2017) [23]	MWe	4–750 MWe; extrapolated below

The Interim Report used ad hoc power-law coefficients for the turbines and approximate terminal temperature differences for the heaters; those were replaced because (i) the coefficients could not be traced to the cited source and (ii) the terminal-difference approximation produced unphysical driving forces for the steam boiler and the sCO_2 heater. Specific capital cost (AUD/kW) is reported at 100 kW, 500 kW, 1 MW and 5 MW net output at each site's design conditions, and the cost breakdown at 1 MWe.

3.7 Techno-Economic Analysis (RQ3)

3.7.1 Case Study Definitions

Three Australian industrial case studies are drawn from ARENA [4] and A2EP [30] data (Table 3.3).

Table 3.3: Australian industrial case study definitions.

Case	Industry / Site	T_{src} (°C)	$\dot{m}_{gas}$ (kg/s)	Price (AUD/MV	Rationale
CS2	Gas processing intercooler (Pilbara, WA)	200	30	130	Representative LNG train; isolated-grid price premium [31]
CS1	Cement preheater exhaust (Gladstone, QLD)	350	50	90	Dominant WHR opportunity in Australian cement
CS3	Electric arc furnace flue gas (Sydney, NSW)	500	20	100	High-temperature WHR in NEM-connected steelmaking

3.7.2 Fixed-Source Basis

The comparison is made on a *fixed-source* basis: each cycle is sized to the net output it can recover from the site's stream (Eq. 3.15), and its capital cost follows from the cost model at that size. This is the decision an operator faces — the stream is given, the plant is not. A fixed-output comparison (all cycles at 1 MWe) is reported in RQ2 for the capital cost; on that basis, because waste heat is free and the plant life and capacity factor are common, LCOE is simply proportional to the specific capital cost, $\text{LCOE} = (\text{CRF} + 0.025)\,\text{TEC}/(8760\,\text{CF})$, and adds no information beyond

RQ2.

3.7.3 LCOE, NPV, Payback and IRR

Levelised Cost of Electricity is calculated following Short et al. [26]:

$$\mathrm{LCOE} = \frac{C_{cap} \cdot \mathrm{CRF} + C_{OM}}{E_{gen}} \tag{3.18}$$

where C_{cap} is the total erected cost (AUD), C_{OM} is annual operations and maintenance (2.5 % of capital per year, consistent with WHR industry practice [14]), E_{gen} is annual electricity generation (MWh/yr at 85 % capacity factor), and CRF is the capital recovery factor at a 7 % real discount rate over a 20-year project life:

$$\mathrm{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1} = 0.0944 \tag{3.19}$$

Net present value is the sum of discounted annual net revenue (electricity revenue minus O&M) over the project life less the initial capital. Payback is reported as *discounted* payback — the year in which cumulative discounted net cash flow first exceeds the capital, interpolated within the year — so that it is consistent with NPV and LCOE; the simple payback is also recorded. Internal rate of return is the discount rate at which NPV is zero. Electricity prices are drawn from AEMO spot market data and WA wholesale tariffs [31] (Table 3.3). Waste heat is free, no carbon price is applied, and the cost of mechanical chilling for the tCO_2 condenser is not included (Section 4.6 quantifies it separately).

3.7.4 Monte Carlo Uncertainty Analysis

Uncertainty is quantified by 10,000-sample Monte Carlo, varying three parameters simultaneously with uniform distributions: capital cost (±30 %, matching the stated uncertainty of the cost correlations), electricity price (±40 %) and capacity factor (60–95 %). The primary output is $P(\mathrm{NPV} > 0)$, with P10/P50/P90 values of NPV and LCOE. A breakeven analysis also reports the reduction in heat exchanger cost (primary heater plus recuperator) at which each CO_2 cycle would match the best cycle's LCOE at the same site.

3.8 Off-Design Analysis (RQ4)

Part-load performance is evaluated at five load levels — 100, 80, 60, 40, and 20 % of design mass flow — using a simplified Stodola efficiency correction:

$$\eta_{od} = \eta_0 \cdot [1 - 0.3(1 - f)^2] \tag{3.20}$$

where f is the part-load mass-flow fraction and the parabolic coefficient 0.3 is calibrated to Yang et al.'s [27] tCO_2 part-load data. The correction is applied to the turbine of every cycle and additionally to the sCO_2 compressor. Sliding-pressure control is applied: $P_{high} = P_{design}\, f$ for the SRC, $P_{high} = 20\,(0.5 + 0.5f)$ MPa for sCO_2 and $P_{high} = P_{design}\,(0.6 + 0.4f)$ for tCO_2, with low-side pressures and turbine inlet temperature held at their site design values. The heater integration is not re-solved at part load.

Operational flexibility is scored using the Marchionni et al. [17] framework with four criteria: turndown ratio (weight 0.30), efficiency retention at 60 % load (0.30), efficiency retention at 40 % load (0.25), and startup complexity (0.15). Retention values are averaged over the three sites. Startup complexity scores are assigned qualitatively based on the literature: SRC 0.55 (steam drum pressurisation, blowdown, long warm-up required); sCO_2 0.75 (no phase change, faster startup, critical control near compressor operating point); tCO_2 0.80 (simplest startup, liquid pump, no near-critical compressor management [17]).

3.9 Implementation and Reproducibility

All models are implemented in a single Python package (`co2whr`) with one script per research question and a driver (`run_all.py`) that regenerates every figure and every number in this thesis in under two minutes (Appendix A). Results are written to a machine-readable file (`results.json`) from which the tables in Chapters 4 to 7 were transcribed.

4. Thermodynamic Performance Results (RQ1)

This chapter presents all results addressing RQ1. Section 4.2 covers first-law parametric performance; Section 4.3 the second-law exergy analysis; Section 4.4 the pseudo-critical heater integration; and Sections 4.5 and 4.6 the recuperator and condensing-pressure sensitivity sweeps, together with the warm-climate sensitivity of the sCO_2 compressor inlet. The thirteen findings of this chapter differ in several respects from the Interim Report because of the recuperator correction described in Section 3.2.2; the differences are stated explicitly where they arise.

4.1 Model Validation

SRC validation against NIST Webbook gave a maximum enthalpy deviation of 0.031 % (Table 3.1), well inside the 1 % criterion. CoolProp's CO_2 critical point ($T_c = 30.98\,°C$, $P_c = 7.377\,MPa$) matches NIST REFPROP to within 0.04 %. Physical self-consistency checks at the SRC baseline (400 °C TIT, 10 MPa, 10 kPa condenser) gave condenser temperature 45.8 °C, BWR 1.31 % and η_{th} 32.79 %, all within expected ranges for subcritical steam at this condition. The global exergy balance (Eq. 3.11) closes to machine precision for all three cycles, and the discretised recuperator shows a positive temperature difference at every node for every case reported (the smallest, 2.7 K, is the sCO_2 recuperator at the 200 °C site, discussed in Section 5.1).

4.2 First-Law Parametric Performance

Figure 4.1 shows thermal efficiency and back-work ratio across 150–600 °C turbine inlet temperature for all three cycles under the sweep rules of Section 3.2, and Table 4.1 extracts the values at the three Australian case study temperatures.

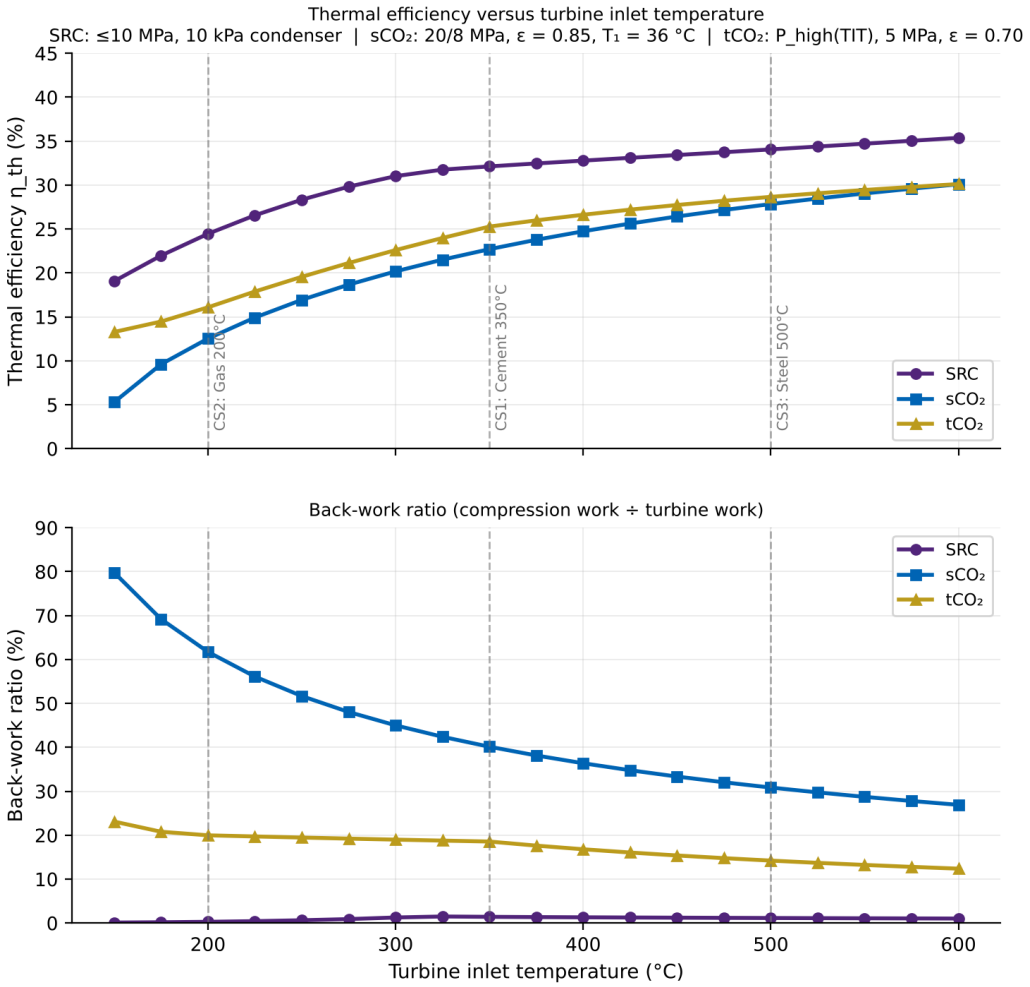


Figure 4.1: Three-cycle thermal efficiency (top) and back-work ratio (bottom) versus turbine inlet temperature, 150–600 °C. Dashed lines mark the Australian case study temperatures. Parameters: $\eta_{turbine} = 0.85$, $\eta_{pump/comp} = 0.80$; SRC: ≤10 MPa, 10 kPa condenser; sCO_2: 20/8 MPa, $\varepsilon = 0.85$, compressor inlet 36 °C; tCO_2: $P_{high}(T)$, $P_{low} = 5$ MPa, $\varepsilon = 0.70$. Recuperators limited by both streams (Eq. 3.5).

Finding 1 — Under the modelled assumptions the steam Rankine cycle has the highest thermal efficiency at every turbine inlet temperature from 150 to 600 °C. SRC rises from 19.06 % to 35.38 %; tCO_2 from 13.28 % to 30.12 %; sCO_2 from 5.29 % to 30.09 %. The two CO_2 cycles converge on each other at 600 °C (**30.1 % for both**) but remain 5.3 percentage points below steam. No crossover occurs within the WHR range. This reverses the Interim Report's Findings 1 and 2 (sCO_2/SRC crossover at 362 °C; tCO_2 ahead of SRC everywhere), both of which rested on recuperators that transferred more heat than the hot stream could supply (Section 3.2.2). The steam figures are unchanged.

Finding 2 — sCO_2 back-work ratio is prohibitive below ≈300 °C. At 150 °C the compressor absorbs **79.8 % of turbine output**; at 200 °C, 61.7 %; at 300 °C, 45.0 %; and still 26.9 % at 600 °C. The tCO_2 pump absorbs 23.1 % at 150 °C (higher than the literature-reported 2–5 % owing to $P_{low} = 5$ MPa), falling to 12.4 % at 600 °C. The SRC feed pump stays below 1.5 % throughout. BWR, not turbine work, is why the CO_2 cycles trail steam: the gross specific turbine work of the CO_2 cycles is comparable to steam per unit of heat input, but a quarter to three-quarters of it is returned to the compressor.

Finding 3 — The recuperator gain is bounded by the hot stream. With the effectiveness applied to the limiting stream, recuperation at $\varepsilon = 0.70$ raises the tCO_2 efficiency by **+3.4, +7.0 and +10.1 percentage points** at the 200, 350 and 500 °C site conditions (12.2→15.7 %, 18.0→25.0 %, 18.4→28.5 %), compared with the +12.3, +16.8 and +20.3 points obtained at the interim stage with a cold-side-only effectiveness. The hot-side limit binds because the 5 MPa exhaust has a much smaller enthalpy capacity than the near-critical high-pressure stream it heats.

Table 4.1: Thermal efficiency and back-work ratio at the Australian case study turbine inlet temperatures (parametric sweep rules; $\varepsilon_{sCO_2} = 0.85$, $\varepsilon_{tCO_2} = 0.70$, $P_{low,tCO_2} = 5$ MPa).

Parameter	200 °C (CS2)	350 °C (CS1)	500 °C (CS3)
SRC η_{th} (%)	24.45	32.13	34.06
sCO_2 η_{th} (%)	12.53	22.70	27.84
tCO_2 η_{th} (%)	16.09	25.28	28.66
SRC BWR (%)	0.29	1.42	1.15
sCO_2 BWR (%)	61.74	40.14	30.85
tCO_2 BWR (%)	19.98	18.56	14.22
Best cycle (η_{th})	SRC	SRC	SRC

4.3 Second-Law Analysis

4.3.1 Basis of Comparison

Exergy destruction is conventionally tabulated per kilogram of working fluid. For a comparison of cycles that is misleading, because at a common turbine inlet temperature the three cycles circulate very different mass flows per kilowatt: at 400 °C the net specific work is 948 kJ/kg for steam, 108 kJ/kg for tCO_2 and 57 kJ/kg for sCO_2. A kilogram of CO_2 destroys far less exergy than a kilogram of steam simply because it does far less work. Figure 4.2 therefore shows both bases, and Table 4.2 reports destruction per kilojoule of net work, which is the like-for-like quantity. The Interim Report's statement that sCO_2 destroys 95 % less exergy than steam compared the per-kilogram totals and is withdrawn.

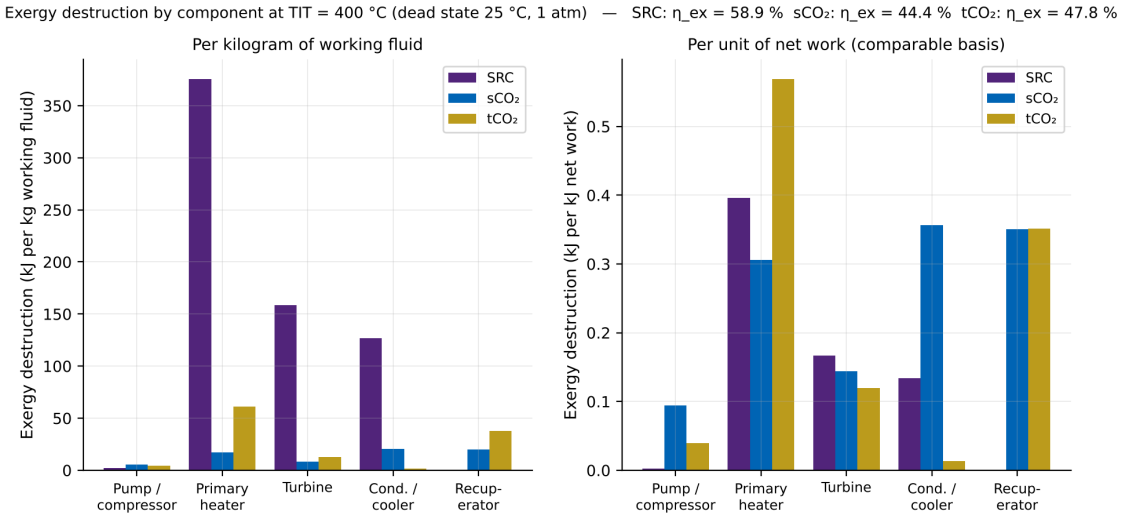


Figure 4.2: Component exergy destruction at a common turbine inlet temperature of 400 °C (dead state 25 °C, 101.325 kPa), per kilogram of working fluid (left) and per kilojoule of net work (right). The right-hand panel is the comparable basis.

Table 4.2: Exergy destruction at 400 °C turbine inlet temperature, in kJ per kJ of net work (per-kilogram totals in the last column).

Cycle	Pump/ comp.	Heater	Turbine	Cond./ cooler	Recup.	Total	η_{ex} (%)	Total (kJ/kg)
SRC	0.002	0.396	0.167	0.134	—	**0.699**	58.9	663.2
sCO_2	0.094	0.306	0.144	0.357	0.351	**1.252**	44.4	71.5
tCO_2	0.040	0.569	0.120	0.013	0.351	**1.092**	47.8	117.7

Finding 4 — The steam boiler is the single largest loss in any cycle, but per unit of work steam remains the most exergy-efficient cycle at 400 °C. The boiler destroys **375 kJ/kg, 56.6 % of all SRC losses**, because water evaporates at a fixed temperature well below the source. Yet the SRC destroys only 0.70 kJ of exergy per kJ of net work against 1.09 for tCO_2 and 1.25 for sCO_2, with exergy efficiencies of 58.9, 47.8 and 44.4 %. The CO_2 cycles' non-isothermal heating does reduce heater destruction per unit of work (0.31 for sCO_2 versus 0.40 for steam), but the saving is more than consumed by their recuperators (0.35 each) and, for sCO_2, by the compressor and gas cooler (0.09 and 0.36).

Finding 5 — For the CO_2 cycles the recuperator, not the heater, is the dominant internal irreversibility. At 400 °C the recuperator accounts for 28 % of sCO_2 destruction and 32 % of tCO_2 destruction, as large as the heater in the sCO_2 case. The large temperature differences that a fixed-effectiveness recuperator tolerates when the two streams have very different c_p are a genuine second-law cost, and they grow with source temperature (Section 4.5).

4.3.2 Exergy at the Site Design Points

Table 4.3 repeats the analysis at the three site design points of Section 5.1, where the steam cycle operates at the pinch-limited boiler pressure rather than at 10 MPa. Two features stand out. First, with a realistic boiler pressure the tCO_2 cycle is the most exergy-efficient option at 200 and 350 °C (42.4 and 47.9 % against 39.0 and 45.4 % for steam), and steam leads only at 500 °C. Second, the tCO_2 condenser term is *negative* at 200 and 350 °C. The condenser rejects heat at 14.3 °C, below the 25 °C dead state, so the condensate leaving it carries more exergy ("cold exergy") than the vapour entering. The balance closes only because the model does not charge the cycle for the refrigeration needed to supply a 14 °C sink; that charge is quantified in Section 4.6.

Table 4.3: Exergy destruction at the site design points (per kJ net work) with the heat source treated as a reservoir at T_{src}.

Site	Cycle	Pump/ comp.	Heater	Turbin	Cond./ cooler	Recup.	Total	η_{th} (%)	η_{ex} (%)
CS2	SRC	0.000	1.010	0.165	0.387	—	1.562	14.4	39.0
CS2	sCO_2	0.298	0.807	0.390	0.724	0.016	2.235	11.4	30.9
CS2	tCO_2	0.052	1.085	0.175	−0.140	0.187	1.360	15.7	42.4
CS1	SRC	0.000	0.828	0.165	0.210	—	1.204	23.7	45.4
CS1	sCO_2	0.114	0.400	0.172	0.386	0.272	1.345	22.2	42.6
CS1	tCO_2	0.046	0.658	0.137	−0.017	0.265	1.089	25.0	47.9
CS3	SRC	0.001	0.681	0.166	0.146	—	0.994	30.8	50.2
CS3	sCO_2	0.075	0.261	0.117	0.335	0.440	1.228	27.6	44.9
CS3	tCO_2	0.033	0.504	0.101	0.055	0.463	1.157	28.5	46.4

4.4 Transitional Effects: Pseudo-Critical Heater Integration

Figures 4.3–4.5 show the tCO_2 heater T-h diagrams at the three site conditions with $\varepsilon = 0$, so that the pseudo-critical c_p peak lies inside the heater, and Table 4.4 summarises the integration.

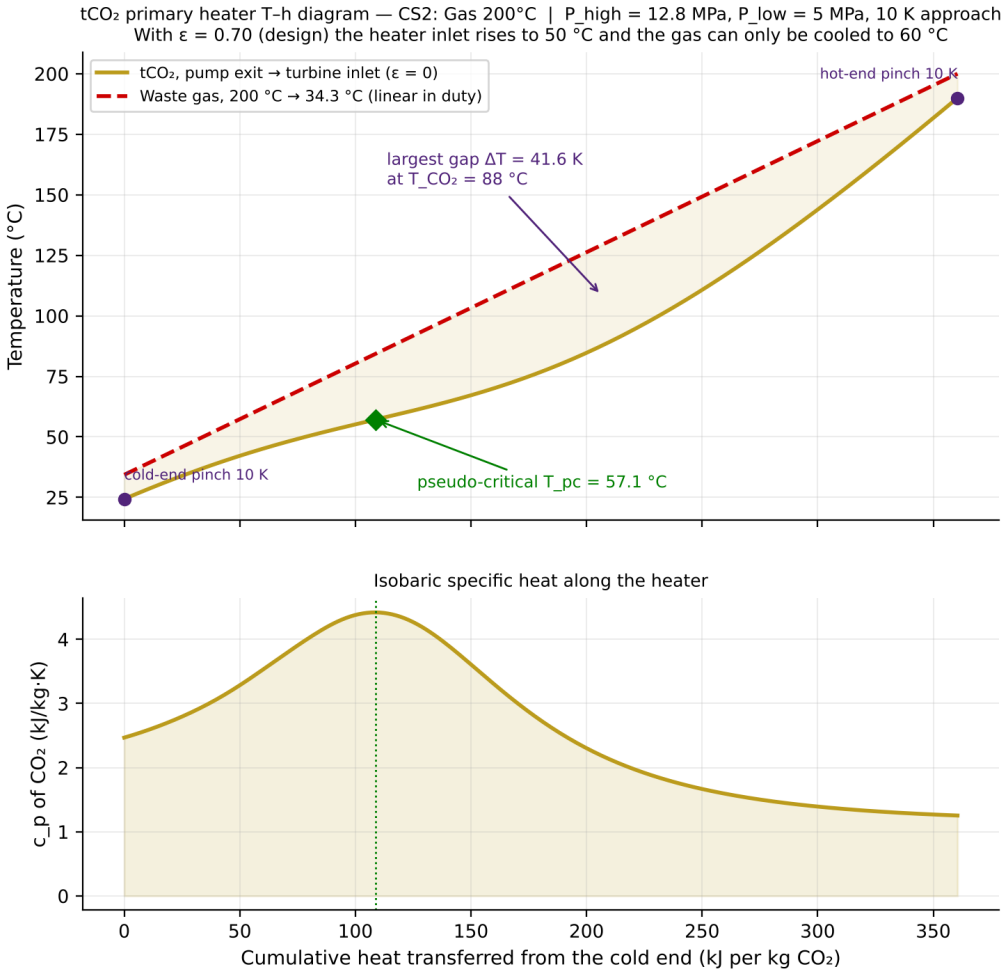


Figure 4.3: tCO_2 heater T-h diagram and c_p profile for CS2 gas processing (200 °C source, 12.8 MPa, $\varepsilon = 0$). Upper panel: CO_2 temperature (gold) and waste-gas temperature (red dashed, linear in heat duty) versus cumulative heat transferred from the cold end. The approach of 10 K is reached at *both* terminals; the gap widens to 41.6 K at a CO_2 temperature of 88 °C, just beyond the pseudo-critical temperature of 57.1 °C. Lower panel: c_p along the heater, peak 4.42 kJ/(kg·K).

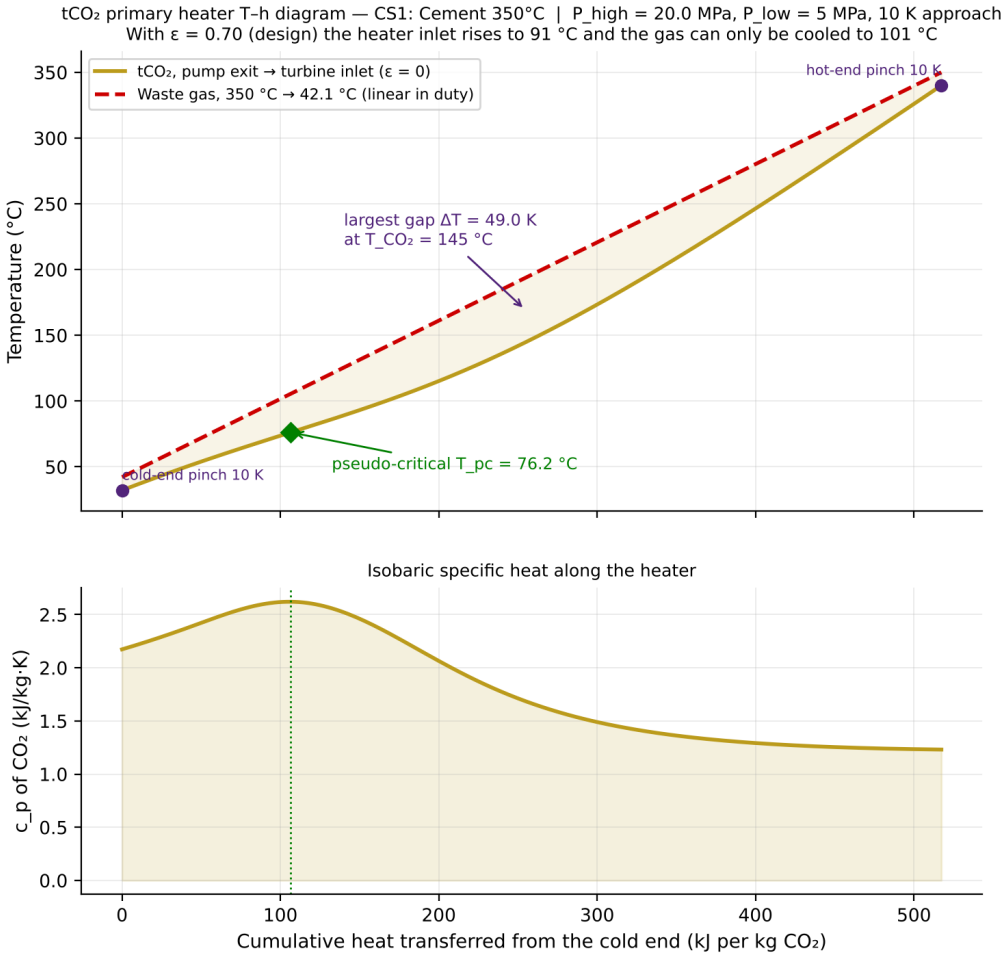


Figure 4.4: tCO_2 heater T-h diagram and c_p profile for CS1 cement (350 °C source, 20.0 MPa, $\varepsilon = 0$). At 20 MPa the c_p peak is broader and lower (2.62 kJ/(kg·K) at 76 °C); the pump exit temperature rises to 32.1 °C, so the gas cannot be cooled below 42.1 °C. Largest gap 49.0 K at 145 °C.

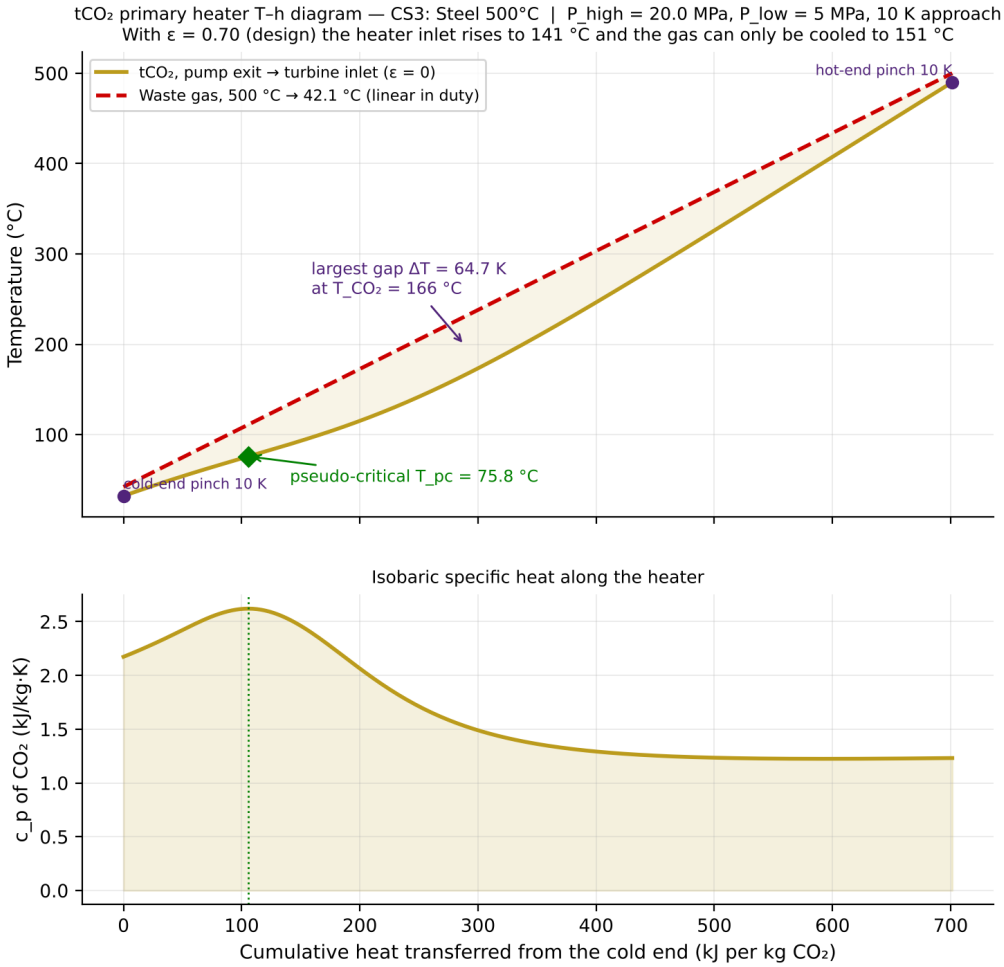


Figure 4.5: tCO_2 heater T-h diagram and c_p profile for CS3 steel (500 °C source, 20.0 MPa, $\varepsilon = 0$). The large source-to-sink span drives total heater duty to 701 kJ/kg; largest gap 64.7 K at 166 °C.

Table 4.4: tCO_2 heater integration at $\varepsilon = 0$ (10 K approach, turbine inlet $T_{src} - 10$ K) and the corresponding design values at $\varepsilon = 0.70$.

Parameter	CS2 (200 °C)	CS1 (350 °C)	CS3 (500 °C)
P_{high} (MPa)	12.8	20.0	20.0
Pump exit temperature (°C)	24.3	32.1	32.1
T_{pc} (°C) / peak c_p (kJ/kg·K)	57.1 / 4.42	76.2 / 2.62	75.8 / 2.62
Gas outlet temperature, $\varepsilon = 0$ (°C)	34.3	42.1	42.1
Heater duty, $\varepsilon = 0$ (kJ/kg)	360.5	517.3	701.3
Minimum ΔT (K) and location	10.0, both ends	10.0, both ends	10.0, both ends
Maximum ΔT (K) at T_{CO_2} (°C)	41.6 at 88	49.0 at 145	64.7 at 166
Effective LMTD, segmented (K)	22.9	27.1	32.3
Heater inlet, $\varepsilon = 0.70$ (°C)	50.2	91.1	141.1
Gas outlet, $\varepsilon = 0.70$ (°C)	60.2	101.1	151.1
Effective LMTD, $\varepsilon = 0.70$ (K)	23.2	21.3	16.1

Finding 6 — The tCO_2 heater is pinched at both terminals, and the pseudo-critical c_p peak opens the driving force in the middle rather than closing it. With the gas profile linear in heat duty, the 10 K approach is reached at the cold end and at the hot end simultaneously, and the temperature difference bulges to **41.6, 49.0 and 64.7 K** at CO_2 temperatures of 88, 145 and 166 °C, downstream of T_{pc}. The mechanism is that CO_2 absorbs a large share of the duty while its temperature is still low (high c_p), so its curve falls below the straight line joining its end points while the gas, with constant c_p, follows that straight line. The Interim Report's Finding 7 ("the pinch sits mid-exchanger at 129 °C") was an artefact of an index-linear gas profile that produced a constant 10 K gap everywhere, and is withdrawn.

Finding 7 — The heater inlet temperature sets a hard floor on waste heat recovery, and recuperation raises that floor. At $\varepsilon = 0$ the floor is the pump exit: the gas cannot leave below **34.3 °C** at 12.8 MPa or 42.1 °C at 20 MPa. At the design effectiveness of 0.70 the heater inlet rises to 50, 91 and 141 °C and the gas can only be cooled to 60, 101 and 151 °C. Recuperation therefore trades depth of heat recovery for cycle efficiency, a trade quantified in Section 4.5.

Finding 8 — The pseudo-critical peak sharpens as pressure falls. At 12.8 MPa, $T_{pc} = 57.1$ °C with peak $c_p = 4.42\,\mathrm{kJ/(kg{\cdot}K)}$; at 20 MPa, $T_{pc} = 76$ °C with peak $c_p = 2.62\,\mathrm{kJ/(kg{\cdot}K)}$. Lower pressure produces the more non-uniform profile and therefore the larger departure between segmented and two-point sizing, but the pump exit temperature is also lower, which deepens heat recovery. The high-side pressure is thus a trade-off between heater non-uniformity and recovery depth, in addition to its effect on efficiency.

4.5 Recuperator Sensitivity

Figure 4.6 and Table 4.5 show the tCO_2 cycle at $\varepsilon = 0$, 0.50, 0.70 and 0.85 at each site design point.

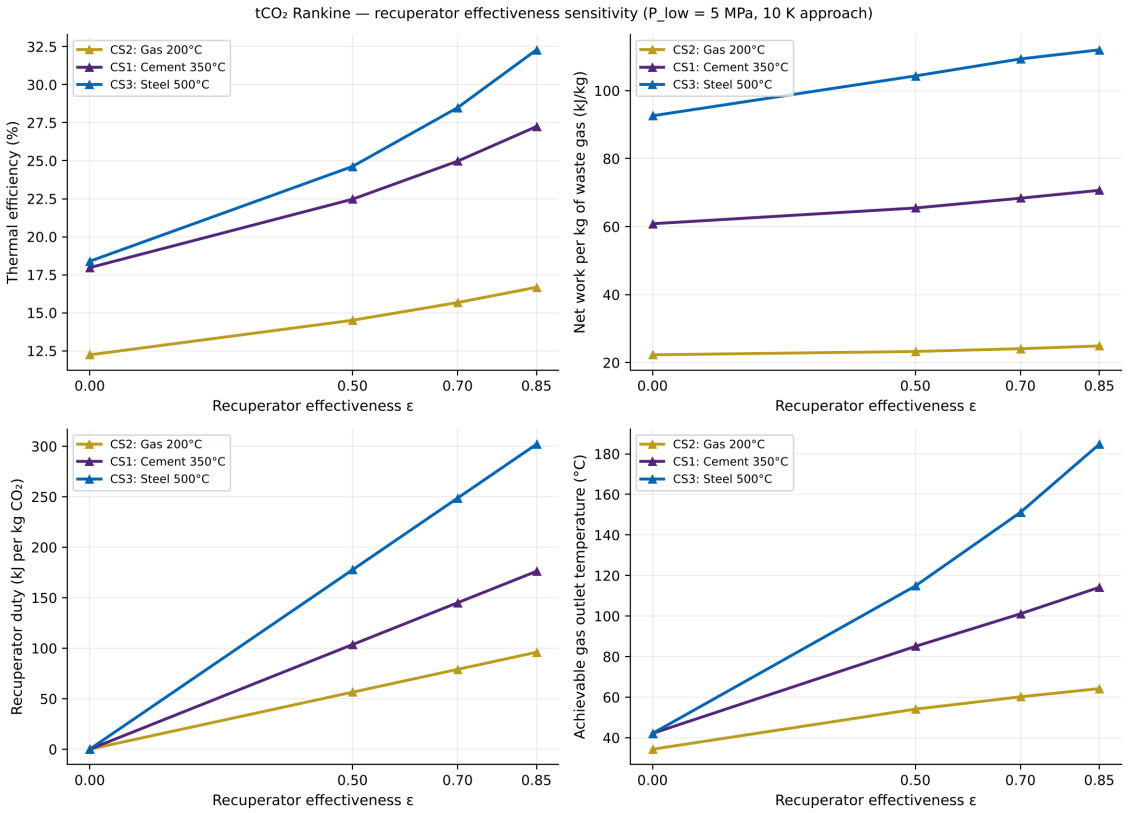


Figure 4.6: tCO_2 recuperator effectiveness sensitivity at the three site design points: thermal efficiency, net work per kilogram of waste gas, recuperator duty and the achievable gas outlet temperature.

Table 4.5: tCO_2 recuperator sensitivity ($P_{low} = 5\,\text{MPa}$, 10 K approach). w_{gas} is the net work per kilogram of waste gas (Eq. 3.15).

Site	ε	η_{th} (%)	BWR (%)	q_{in} (kJ/kg)	Q_{recup} (kJ/kg)	Heater inlet (°C)	Gas out (°C)	w_{gas} (kJ/kg gas)
CS2	0	12.23	20.69	360.5	0.0	24.3	34.3	22.30
CS2	0.50	14.51	20.69	304.0	56.5	44.1	54.1	23.28
CS2	0.70	15.67	20.69	281.4	79.1	50.2	60.2	24.10
CS2	0.85	16.68	20.69	264.5	96.0	54.2	64.2	24.91
CS1	0	17.97	18.97	517.3	0.0	32.1	42.1	60.85
CS1	0.50	22.47	18.97	413.7	103.6	75.0	85.0	65.49
CS1	0.70	24.97	18.97	372.3	145.1	91.1	101.1	68.36
CS1	0.85	27.25	18.97	341.2	176.2	104.2	114.2	70.68
CS3	0	18.39	14.44	701.3	0.0	32.1	42.1	92.60
CS3	0.50	24.62	14.44	523.7	177.6	104.8	114.8	104.33
CS3	0.70	28.49	14.44	452.7	248.7	141.1	151.1	109.32
CS3	0.85	32.29	14.44	399.4	302.0	174.7	184.7	111.99

Finding 9 — Recuperation improves efficiency far more than it improves recovered work, because every increment of ε raises the gas outlet temperature. From $\varepsilon = 0$ to 0.85, η_{th} rises by 4.5, 9.3 and 13.9 points at CS2, CS1 and CS3, but w_{gas} rises by only **+12 %, +16 % and +21 %**, because the gas outlet temperature climbs from 34–42 °C to 64–185 °C. The marginal benefit also diminishes: the step 0.70→0.85 adds 0.8–2.7 kJ per kilogram of gas while roughly doubling the recuperator conductance. BWR is independent of ε (20.7, 19.0 and 14.4 %), confirming that recuperation reduces external heat input only and does not alter the pump or turbine work.

Finding 10 — At high effectiveness the recuperator approaches the primary heater in duty, and at low source temperature it approaches the 10 K approach limit. At $\varepsilon = 0.85$ the recuperator transfers 302 kJ/kg at CS3, 76 % of the external heat input (399 kJ/kg). At CS2 the same effectiveness leaves a minimum internal temperature difference of only **9.8 K**, essentially at the design approach, so $\varepsilon = 0.85$ is the practical ceiling for a 200 °C source at 12.8 MPa. The baseline of 0.70 retains 21–86 K of internal margin at all three

sites.

4.6 Low-Side Pressure and Heat-Sink Sensitivity

Figure 4.7 and Table 4.6 show the tCO_2 cycle at low-side pressures from 3.5 to 7.0 MPa with $\varepsilon = 0.70$.

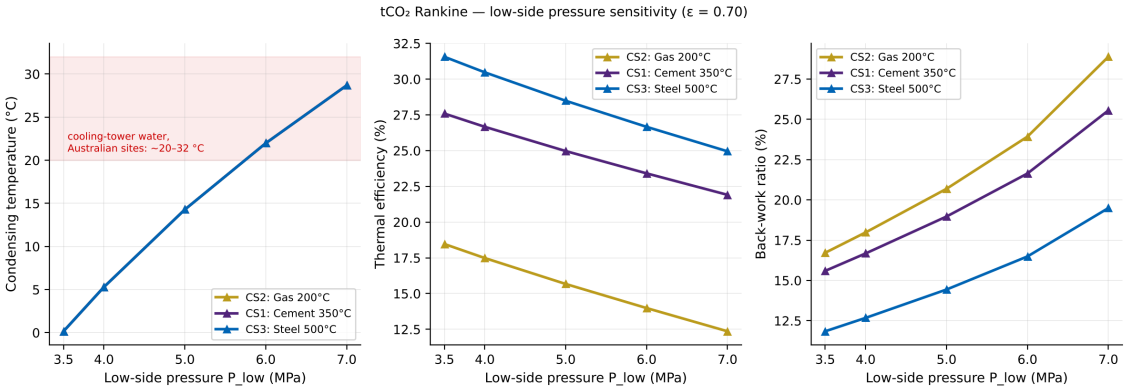


Figure 4.7: tCO_2 low-side pressure sensitivity ($\varepsilon = 0.70$): condensing temperature, thermal efficiency and back-work ratio. The shaded band marks the cooling-tower water temperatures typical of the three Australian sites.

Table 4.6: tCO_2 low-side pressure sensitivity ($\varepsilon = 0.70$, 10 K approach).

Site	P_{low} (MPa)	T_{cond} (°C)	η_{th} (%)	BWR (%)	w_{net} (kJ/kg)	Gas out (°C)	w_{gas} (kJ/kg gas)
CS2	3.5	0.2	18.47	16.72	61.6	44.7	31.56
CS2	4.0	5.3	17.49	17.98	55.1	50.7	28.72
CS2	5.0	14.3	15.67	20.69	44.1	60.2	24.10
CS2	6.0	22.0	13.99	23.93	34.8	67.8	20.33
CS2	7.0	28.7	12.36	28.88	26.3	76.5	16.80
CS1	3.5	0.2	27.59	15.59	117.5	80.4	81.82
CS1	4.0	5.3	26.67	16.68	108.5	87.6	76.96
CS1	5.0	14.3	24.97	18.97	93.0	101.1	68.36
CS1	6.0	22.0	23.41	21.64	79.5	114.8	60.57
CS1	7.0	28.7	21.91	25.55	67.0	131.0	52.79
CS3	3.5	0.2	31.56	11.84	161.5	120.2	131.86
CS3	4.0	5.3	30.48	12.68	149.6	130.4	123.92
CS3	5.0	14.3	28.49	14.44	129.0	151.1	109.32
CS3	6.0	22.0	26.67	16.49	111.2	172.9	95.93
CS3	7.0	28.7	24.95	19.49	94.9	197.7	82.97

Finding 11 — The thermodynamically attractive condensing pressures are climatically infeasible in Australia, and the 5 MPa baseline is itself marginal. $P_{low} = 3.5$ MPa condenses at 0.2 °C and 4.0 MPa at 5.3 °C — unattainable at any of the three sites. The baseline 5 MPa condenses at **14.3 °C**, which requires a sink near 4 °C for a 10 K approach: colder than cooling-tower water anywhere in Australia (typically 20–32 °C [33]) and achievable only with mechanical chilling. Only $P_{low} \geq 6.5$ MPa (condensing above 25 °C) is compatible with evaporative cooling at Gladstone or Sydney outside summer, and no subcritical pressure is compatible with Pilbara summer conditions.

Finding 12 — The cost of a cold sink would exceed the benefit of the cycle at low source temperatures. If the 14.3 °C condenser at 5 MPa were served by an electric chiller, the parasitic load would be q_{out}/COP. With $q_{out} = 237$, 279 and 324 kJ/kg and $w_{net} = 44$, 93 and 129 kJ/kg, a chiller of

COP 4 would consume **134 %, 75 % and 63 %** of the net output at CS2, CS1 and CS3 (90, 50 and 42 % at COP 6). Raising P_{low} to 7 MPa (28.7 °C) removes the need for chilling at the cost of 3.1–3.5 efficiency points and 23–30 % of w_{gas}; at CS1 the resulting 52.8 kJ per kilogram of gas equals that of the pinch-limited steam cycle (Section 5.1).

Finding 13 — The warm-climate penalty on sCO_2 is concentrated at low source temperatures. Raising the compressor inlet from 32 to 40 °C (8 MPa) reduces sCO_2 efficiency from **15.7 to 9.2 %** at a 200 °C turbine inlet (a 42 % relative loss), from 23.6 to 21.3 % at 350 °C, and from 28.0 to 27.1 % at 500 °C. The baseline of 36 °C, chosen as 5 K above the critical temperature to guarantee single-phase compression with Australian cooling water, gives 12.5, 22.7 and 27.8 %. Near-critical compression is the source of the sCO_2 cycle's advantage in the literature and it is exactly what a warm heat sink takes away.

4.7 Comparison with Published Results

Kizilkan [13] reported 27.6 % for sCO_2 and 24.2 % for steam at a 350 °C cement source. The present steam value at the 350 °C *site design point* (pinch-limited to 0.91 MPa, Section 5.1) is 23.7 %, in close agreement; the present sCO_2 value (22.2 % at the site, 22.7 % in the sweep) is lower than Kizilkan's because his configuration is a recompression cycle with a compressor inlet near 32 °C, whereas this study models a simple recuperated layout with a 36 °C inlet and a hot-side-limited recuperator. Jiang et al. [11] found tCO_2 up to 5 points ahead of sCO_2; this study finds tCO_2 ahead of sCO_2 by 2.6–3.6 points between 200 and 350 °C, narrowing to 0.8 points at 500 °C, consistent in direction. Crespi et al.'s [9] 40–50 % efficiencies for sCO_2 apply above 500 °C with optimised recompression layouts and cold sinks, conditions outside the WHR range and the Australian climate considered here. The disagreement with the literature is therefore not with the thermodynamics but with the boundary conditions, and Findings 11–13 identify which ones.

4.8 Summary of RQ1 Results

Under consistent assumptions and a second-law-consistent recuperator, the steam Rankine cycle has the highest thermal efficiency at every turbine inlet temperature in the 150–600 °C range; tCO_2 is second below 500 °C and the two CO_2 cycles converge above it. Per unit of net work the SRC is also the most exergy-efficient cycle

at a common turbine inlet temperature, although at realistic pinch-limited boiler pressures the tCO_2 cycle leads at 200 and 350 °C. The tCO_2 heater is pinched at both terminals with the driving force widest in the middle, so the transitional effect enlarges rather than restricts the mean temperature difference. Recuperation raises efficiency but also the gas outlet temperature, so it adds far less to the work recovered from a stream than to the efficiency. Finally, the CO_2 cycles' sensitivity to the heat sink is their defining weakness in Australia: the tCO_2 baseline needs a 14 °C condenser that would cost more to provide than the cycle produces at 200 °C, and the sCO_2 cycle loses two-fifths of its efficiency at 200 °C if its compressor inlet warms from 32 to 40 °C.

5. Equipment Sizing and Capital Cost (RQ2)

This chapter answers RQ2 in four steps. Section 5.1 establishes the design point of each cycle at each site, including the pinch-limited steam pressure and the power each cycle can recover from the stream. Section 5.2 compares turbine compactness, Section 5.3 compares heat exchanger sizing methods, and Section 5.4 reports capital cost versus plant size and its breakdown by component.

5.1 Site Design Points

Table 5.1 gives the design point of every cycle at every site after integration with the waste-gas stream (Section 3.6.1). Two numbers in each row matter more than the efficiency: the gas outlet temperature, which fixes how much of the stream is used, and w_{gas}, the net work obtained per kilogram of gas.

Table 5.1: Site design points (turbine inlet $T_{src} - 10\,\mathrm{K}$, 10 K heater approach). T^* is the saturation temperature of the steam boiler (s), the sCO_2 compressor inlet temperature (i) or the tCO_2 condensing temperature (c). w_{gas} is the net work per kilogram of waste gas and $\dot{W}_{site}$ the net output from the whole stream of Table 3.3.

Site	Cycle	P_{high} (MPa)	T^* (°C)	η_{th} (%)	BWR (%)	Heater in (°C)	Gas out (°C)	w_{gas} (kJ/kg)	$\dot{W}_{site}$ (kW)
CS2	SRC	0.16	113.1 s	14.44	0.0	45.8	113.6	13.7	412
CS2	sCO_2	20.0	36.0 i	11.43	64.4	100.0	110.0	11.3	340
CS2	tCO_2	12.8	14.3 c	15.67	20.7	50.2	60.2	24.1	**723**
CS1	SRC	0.91	175.9 s	23.67	0.2	45.9	147.2	52.8	2639
CS1	sCO_2	20.0	36.0 i	22.24	41.0	177.5	187.5	39.8	1988
CS1	tCO_2	20.0	14.3 c	24.97	19.0	91.1	101.1	68.4	**3418**
CS3	SRC	3.69	245.7 s	30.82	0.5	46.2	164.7	113.7	**2273**
CS3	sCO_2	20.0	36.0 i	27.58	31.3	279.0	289.1	64.0	1280
CS3	tCO_2	20.0	14.3 c	28.49	14.4	141.1	151.1	109.3	2186

Finding 14 — Integrated with a real stream, the steam cycle must run at a far lower pressure than the parametric sweep suggests. The boiler pinch at the start of evaporation limits the optimum pressure to **0.16,**

0.91 and 3.69 MPa at the 200, 350 and 500 °C sites (saturation 113, 176 and 246 °C), giving site efficiencies of 14.4, 23.7 and 30.8 % against 24.5, 32.1 and 34.1 % in the sweep. A higher pressure raises efficiency but forces the gas to leave hotter, and the product $\eta_{th}(T_{src} - T_{gas,out})$ peaks at these values. Any comparison that evaluates steam at 10 MPa against a 200–350 °C exhaust stream overstates it badly.

Finding 15 — On a fixed-source basis the tCO_2 cycle recovers the most power at 200 and 350 °C, and the sCO_2 cycle the least at every site. From the CS2 stream tCO_2 delivers **723 kW against 412 kW for steam and 340 kW for sCO_2**; from CS1, 3418 against 2639 and 1988 kW; from CS3, 2186 against 2273 and 1280 kW. The ranking follows the gas outlet temperature, not the efficiency: tCO_2 cools the gas to 60–151 °C because its heater inlet is the recuperator outlet at 50–141 °C; steam cools it to 114–165 °C, limited by the evaporator pinch; the recuperated sCO_2 cycle cools it only to 110–289 °C because its heater inlet is 100–279 °C. At CS3 the sCO_2 cycle uses only 63 % of the heat that steam uses. The sCO_2 recuperator at CS2 also reaches an internal temperature difference of 2.7 K, below the 10 K design approach, so that design point is not practically realisable without a lower effectiveness.

5.2 Turbine Compactness: Size Parameter and Volumetric Flow Ratio

Figure 5.1 and Table 5.2 compare turbine size parameter and volumetric flow ratio at the site design points.

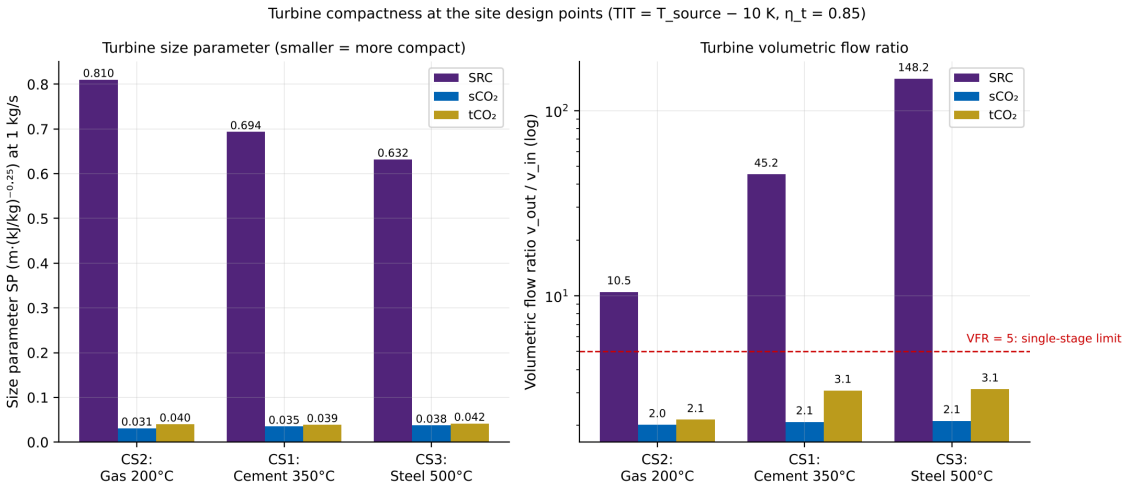


Figure 5.1: Turbine size parameter (left) and volumetric flow ratio (right, log scale) at the site design points. VFR < 5 permits a single-stage radial turbine.

Table 5.2: Turbine size parameter (per kg/s) and volumetric flow ratio at the site design points.

Parameter	CS2 (200 °C)	CS1 (350 °C)	CS3 (500 °C)
SP, SRC ($\mathrm{m{\cdot}(kJ/kg)^{-0.25}}$)	0.810	0.694	0.632
SP, sCO_2	0.031	0.035	0.038
SP, tCO_2	0.040	0.039	0.042
SP ratio SRC/sCO_2, SRC/tCO_2	26, 20	20, 18	17, 15
VFR, SRC	10.5	45.2	148
VFR, sCO_2	2.01	2.08	2.11
VFR, tCO_2	2.14	3.08	3.13

Finding 16 — CO_2 turbines are 15–26 times more compact than the steam turbine at the same site. The size parameter of the steam turbine is 0.63–0.81 $\mathrm{m{\cdot}(kJ/kg)^{-0.25}}$ per kg/s against 0.031–0.042 for the CO_2 cycles; the ratio is largest at the lowest source temperature (**26× at CS2**) because the

low-pressure steam cycle there exhausts a very large specific volume. sCO_2 is 11–29 % more compact than tCO_2, a gap that narrows with temperature.

Finding 17 — The steam turbine needs a multi-stage machine at 350 °C and above; both CO_2 cycles stay single-stage. Steam VFR is 10.5, 45 and 148 at the three site pressures (lower than the 89–807 of the 10 MPa sweep because the site pressures are lower), all above the single-stage threshold of 5. CO_2 VFR is **2.0–3.1** at every site. Compactness is the CO_2 cycles' most robust advantage; it survives every other correction made in this thesis.

5.3 Heat Exchanger Sizing: Segmented versus Two-Point LMTD

Figure 5.2 compares the tCO_2 heater area from the segmented integration (Eq. 3.14) with the two-point LMTD result at $\varepsilon = 0$, using $U = 380\,\mathrm{W/m^2K}$.

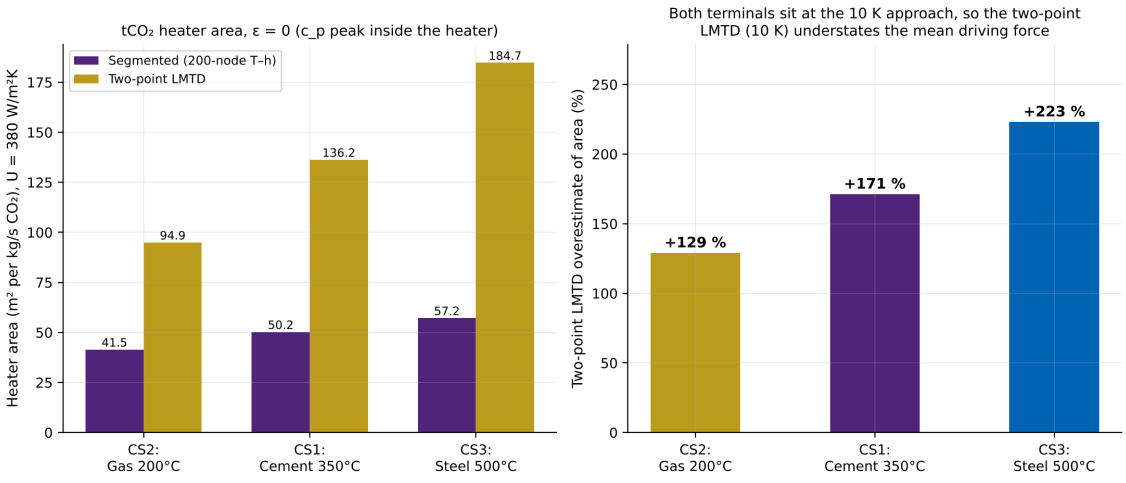


Figure 5.2: tCO_2 heater area at $\varepsilon = 0$: segmented 200-node integration versus two-point LMTD (left), and the percentage by which the two-point method overestimates the area (right).

Finding 18 — Two-point LMTD overestimates the tCO_2 heater area by 129–223 % when the pseudo-critical peak lies inside the heater. Segmented areas are 41.5, 50.2 and 57.2 m^2 per kg/s against 94.9, 136.2 and 184.7 m^2 from the two-point method (**+129, +171 and +223 %**). Because both terminals sit at the 10 K approach (Finding 6), the two-point LMTD is exactly 10 K, whereas the duty-weighted mean driving force is 22.9, 27.1 and 32.3 K. The error grows with source temperature because the mid-exchanger

bulge grows. A designer using the two-point method would price the tCO_2 heater at two to three times its true cost. At the design effectiveness of 0.70 the heater inlet lies above T_{pc}, the profile is nearly linear and the two methods agree to within a few per cent; the capital costs that follow use the segmented UA in all cases.

5.4 Capital Cost Estimation

Figure 5.3 shows specific capital cost against plant size at each site's design conditions, and Figure 5.4 and Table 5.4 the breakdown at 1 MWe. All three cycles are costed with the correlations of Table 3.2.

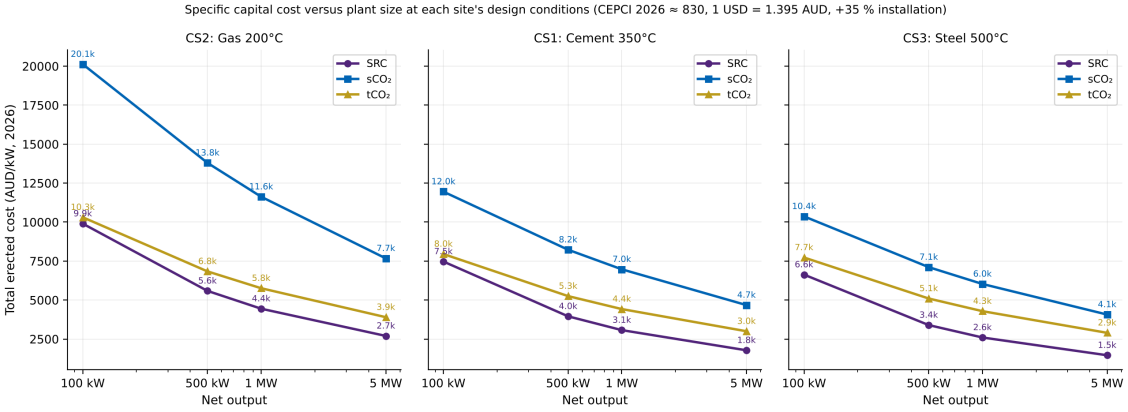


Figure 5.3: Total erected cost per kilowatt (2026 AUD) versus net output at the three site design conditions.

Table 5.3: Specific capital cost (total erected, AUD/kW, 2026) versus net output.

Site	Cycle	100 kW	500 kW	1 MW	5 MW
CS2	SRC	9,900	5,587	**4,444**	2,698
CS2	sCO_2	20,126	13,798	11,626	7,670
CS2	tCO_2	10,308	6,838	5,757	3,899
CS1	SRC	7,469	3,956	**3,071**	1,776
CS1	sCO_2	11,953	8,226	6,971	4,676
CS1	tCO_2	7,951	5,253	4,421	2,998
CS3	SRC	6,630	3,394	**2,598**	1,459
CS3	sCO_2	10,366	7,110	6,031	4,069
CS3	tCO_2	7,728	5,098	4,286	2,897

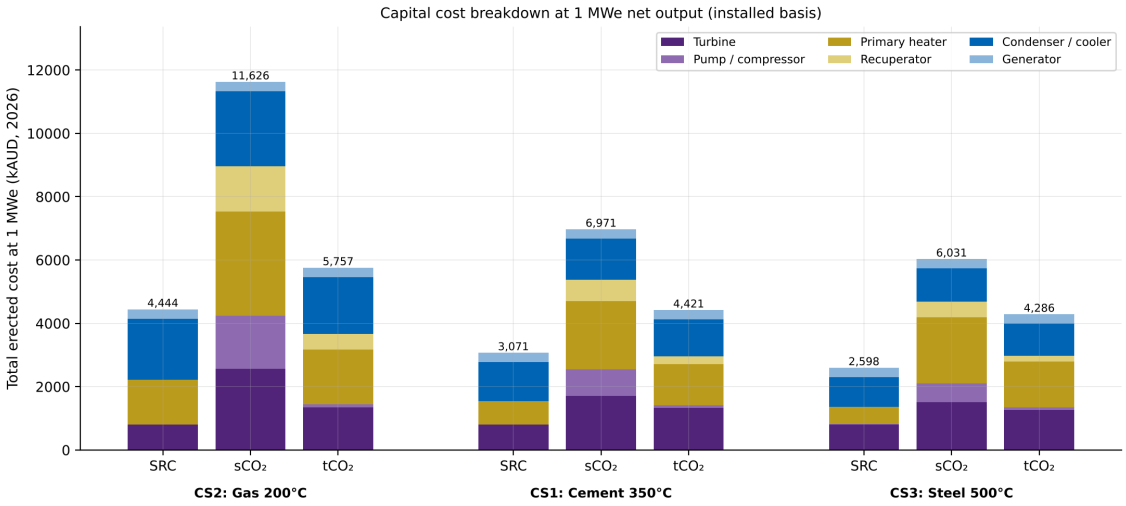


Figure 5.4: Capital cost breakdown at 1 MWe net output (installed basis, 2026 AUD).

Table 5.4: Equipment cost breakdown at 1 MWe (kAUD, 2026, before the 35 % installation factor) and the turbine and compression shaft powers behind it.

Site	Cycle	Turbine	Pump/ comp.	Heater	Recup.	Cooler	Gen.	**Total**	Shaft kW (turb./comp.)
CS2	SRC	593	7	1037	—	1433	222	3292	1000 / 0
CS2	sCO_2	1896	1250	2433	1059	1752	222	8612	2813 / 1813
CS2	tCO_2	998	76	1280	356	1333	222	4265	1261 / 261
CS1	SRC	593	7	544	—	908	222	2275	1002 / 2
CS1	sCO_2	1264	619	1605	489	964	222	5164	1695 / 695
CS1	tCO_2	981	70	960	181	861	222	3275	1234 / 234
CS3	SRC	594	9	408	—	692	222	1924	1005 / 5
CS3	sCO_2	1119	442	1542	363	778	222	4467	1456 / 456
CS3	tCO_2	939	54	1073	135	752	222	3175	1169 / 169

Finding 19 — At equal output the steam cycle is the cheapest to build at every site, tCO_2 costs 30–65 % more and sCO_2 two to three times more. At 1 MWe the total erected cost is **AUD 4,444, 3,071 and 2,598/kW** for steam at CS2, CS1 and CS3; 5,757, 4,421 and 4,286/kW for tCO_2; and 11,626, 6,971 and 6,031/kW for sCO_2. Specific cost falls by 61–78 % between 100 kW and 5 MW, and the ordering is the same at every size. The Interim Report's result that sCO_2 was the cheapest cycle at 200 and 350 °C arose from a sign error in its heater temperature difference that priced the sCO_2 heater at a few per cent of its true cost, and is withdrawn.

Finding 20 — The sCO_2 cost penalty is turbomachinery: a high back-work ratio means every kilowatt of net output needs several kilowatts of machinery. At CS2 a 1 MWe sCO_2 plant needs a **2.8 MW turbine and a 1.8 MW compressor**; turbomachinery is 37 % of its equipment cost against 25 % for tCO_2 and 18 % for steam. The CO_2 turbine is also costed on a cor-

relation (Weiland et al.) whose data begin at 8 MW and which gives a higher unit cost than the steam benchmark at 1 MW (USD 594/kW against 425/kW), reflecting the immaturity of small high-pressure CO_2 machines.

Finding 21 — Heat exchangers are 35–43 % of CO_2 equipment cost, and heat rejection is a large cost for every cycle at low source temperature. The primary heater plus recuperator account for 35–43 % of CO_2 equipment cost and 21–32 % of the steam cycle's. The condenser or cooler, omitted at the interim stage, is **17–44 %** of equipment cost: largest for the 14 %-efficient steam cycle at CS2, which rejects six units of heat per unit of electricity, and smallest for the sCO_2 cycle at CS3. A low-grade WHR plant is, in capital terms, mostly heat exchangers.

5.5 Summary of RQ2 Results

Integrating each cycle with the actual waste-gas stream changes the picture drawn by the parametric sweep. The steam cycle must run at 0.16–3.7 MPa rather than 10 MPa, which lowers its efficiency to 14–31 % but still leaves it the cheapest plant per kilowatt at every site, 30–65 % below tCO_2 and a factor of two to three below sCO_2. The tCO_2 cycle, however, recovers 29 % more power than steam from the 350 °C stream and 75 % more from the 200 °C stream, because its cold heater inlet lets it cool the gas further. The sCO_2 cycle is penalised twice: its hot recuperated heater inlet leaves most of the stream unused, and its back-work ratio makes its turbomachinery the most expensive of the three. The CO_2 cycles' compactness advantage (15–26× in size parameter) is unaffected by any of these corrections. Which of these effects dominates the economics is the subject of RQ3.

6. Techno-Economic Analysis (RQ3)

This chapter answers RQ3 on the fixed-source basis of Section 3.7: each cycle is sized to the output it can recover from the site stream (Table 5.1), costed at that size, and evaluated at the site electricity price. Section 6.1 gives the deterministic results, Section 6.2 examines why the ranking differs from the efficiency ranking, Section 6.3 the Monte Carlo robustness and breakeven analysis.

6.1 Deterministic LCOE, NPV, Payback and IRR

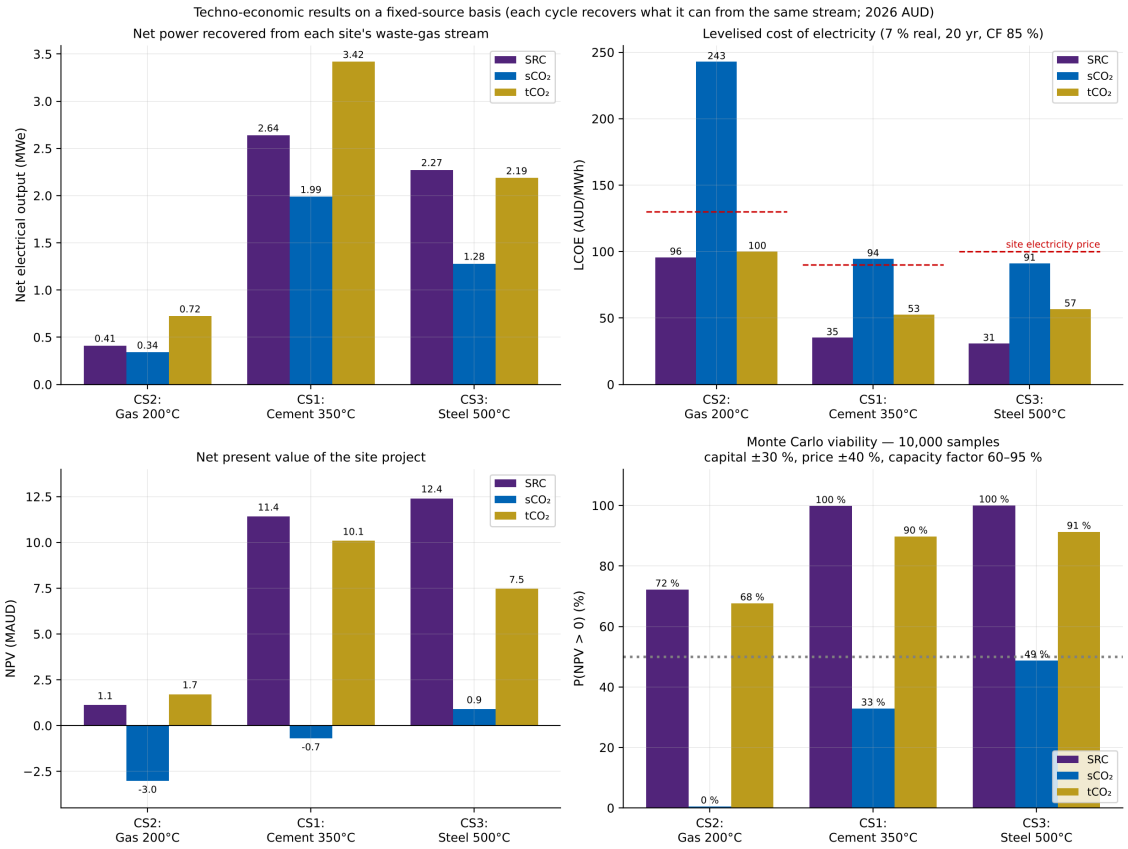


Figure 6.1: Techno-economic results on a fixed-source basis (2026 AUD): net power recovered from each stream, LCOE against the site electricity price, project NPV, and Monte Carlo probability of positive NPV.

Table 6.1: Deterministic techno-economic results on a fixed-source basis (7 % real discount rate, 20-year life, 85 % capacity factor, O&M 2.5 % of capital). Payback is discounted.

Site	Cycle	Output (kW)	TEC (kAUD)	TEC (AUD/kW	LCOE (AUD/MV	NPV (kAUD)	Payback (yr)	IRR (%)
CS2	SRC	412	2,457	5,967	**95.7**	1,115	10.6	12.4
CS2	sCO_2	340	5,144	15,146	242.9	−3,023	never	−2.3
CS2	tCO_2	723	4,511	6,238	100.0	**1,710**	11.4	11.6
CS1	SRC	2639	5,790	2,194	**35.2**	**11,415**	4.2	27.8
CS1	sCO_2	1988	11,715	5,894	94.5	−706	never	6.2
CS1	tCO_2	3418	11,214	3,281	52.6	10,083	7.3	17.2
CS3	SRC	2273	4,368	1,922	**30.8**	**12,406**	3.2	36.2
CS3	sCO_2	1280	7,274	5,684	91.1	895	16.0	8.5
CS3	tCO_2	2186	7,733	3,537	56.7	7,467	7.0	17.9

Finding 22 — The steam cycle gives the lowest cost of electricity at every site. LCOE is **AUD 96, 35 and 31/MWh** for steam at CS2, CS1 and CS3, against 100, 53 and 57/MWh for tCO_2 and 243, 94 and 91/MWh for sCO_2. At the two hotter sites every cycle except sCO_2 generates well below the site electricity price; at the 200 °C Pilbara site steam and tCO_2 sit 23–26 % below the AUD 130/MWh price and sCO_2 is almost twice it. The Interim Report's values of AUD 13–42/MWh for the best cycles were lower because they omitted heat rejection, mis-sized the heaters and assumed a 1 MWe plant regardless of the stream.

Finding 23 — On NPV the tCO_2 cycle is the better investment at 200 °C, because it recovers 75 % more power from the same stream; steam is the better investment at 350 and 500 °C. At CS2 the tCO_2 project is worth **AUD 1.71 M against 1.12 M** for steam, despite a 4 % higher LCOE, because 723 kW earns more than 412 kW. At CS1 steam leads (11.4 M against 10.1 M) even though tCO_2 recovers 29 % more power, because its plant costs 94 % more; at CS3 steam leads clearly (12.4 M against 7.5 M). The sCO_2 project has negative NPV at CS2 and CS1 and is marginal at CS3.

Finding 24 — Discounted payback is 3–4 years for steam at the two hotter sites and 7 years for tCO_2; the 200 °C site pays back in 10–11 years for either. Steam returns its capital in **4.2 and 3.2 years** at CS1 and CS3 with IRRs of 28 and 36 %; tCO_2 in 7.3 and 7.0 years at 17–18 %. At CS2 both are 10.6–11.4 years at 12 % IRR, only five points above the discount rate. The sCO_2 cycle never recovers its capital at CS2 or CS1 and takes 16 years at CS3. (Simple payback, reported at the interim stage, is 2.8–9.4 years for the cases with positive NPV and understates the time to recover capital by 1–7 years.)

6.2 Efficiency versus Recovery Depth

Table 6.2: Why the economic ranking differs from the efficiency ranking: recovered power is the product of efficiency and the depth of heat recovery.

Site	Cycle	η_{th} rank	Gas out (°C)	Heat recovered (kW)	Output (kW)	LCOE rank	NPV rank
CS2	SRC	2	113.6	2,853	412	1	2
CS2	sCO_2	3	110.0	2,970	340	3	3
CS2	tCO_2	1	60.2	4,614	723	2	1
CS1	SRC	2	147.2	11,153	2639	1	1
CS1	sCO_2	3	187.5	8,937	1988	3	3
CS1	tCO_2	1	101.1	13,689	3418	2	2
CS3	SRC	1	164.7	7,376	2273	1	1
CS3	sCO_2	3	289.1	4,641	1280	3	3
CS3	tCO_2	2	151.1	7,675	2186	2	2

Finding 25 — In waste heat recovery, depth of recovery matters as much as efficiency, and the economics follow their product. At CS1 the sCO_2 cycle is only 1.4 efficiency points below steam (22.2 against 23.7 %) yet produces **25 % less power**, because its recuperated heater inlet of 178 °C leaves the gas at 188 °C while steam cools it to 147 °C. At CS2 the tCO_2 cycle is only 1.2 points above steam in efficiency yet produces 75 % more power, because it cools the gas to 60 °C against 114 °C. The useful figure of merit for a WHR cycle is the net work per kilogram of waste gas (Eq. 3.15), and it is on this quantity,

divided by the cost of the plant that delivers it, that the site rankings turn. A fixed-output comparison hides this entirely: at a common 1 MWe, LCOE is simply 0.01604× TEC (AUD/kW) at these economic parameters, and would reproduce the capital-cost ordering of Finding 19 at every site.

6.3 Monte Carlo Uncertainty and Breakeven Analysis

Table 6.3: Monte Carlo results (10,000 samples; capital ±30 %, price ±40 %, capacity factor 60–95 %) and the reduction in heat exchanger cost (heater plus recuperator) at which a cycle would match the best LCOE at its site.

Site	Cycle	P(NPV > 0) (%)	NPV P10 / P50 / P90 (kAUD)	LCOE P10 / P50 / P90 (AUD/MWh)	Determ. NPV (kAUD)	HX cost cut to match best
CS2	SRC	72.2	−723 / 691 / 2,386	77 / 104 / 140	1,115	— (best)
CS2	sCO_2	0.5	−5,191 / −3,310 / −1,423	194 / 265 / 355	−3,023	unreachable
CS2	tCO_2	67.7	−1,541 / 966 / 3,957	80 / 109 / 146	1,710	11 %
CS1	SRC	99.8	3,800 / 9,488 / 16,555	28 / 38 / 51	11,415	— (best)
CS1	sCO_2	32.9	−7,480 / −2,028 / 3,987	76 / 103 / 138	−706	unreachable
CS1	tCO_2	89.7	−81 / 7,623 / 17,108	42 / 57 / 77	10,083	95 %
CS3	SRC	100.0	5,208 / 10,568 / 17,247	25 / 34 / 45	12,406	— (best)
CS3	sCO_2	48.8	−3,745 / −87 / 4,079	73 / 99 / 133	895	unreachable
CS3	tCO_2	91.3	261 / 5,699 / 12,434	45 / 62 / 83	7,467	unreachable

The Monte Carlo medians lie below the deterministic values because the capacity-factor distribution is centred on 77.5 % rather than the 85 % baseline; the P10–P90 spread is dominated by the ±40 % price range.

Finding 26 — Steam at 350 and 500 °C is robust to every combination of cost overrun, price fall and reduced running hours; tCO_2 at those sites is robust in nine cases out of ten; the 200 °C site is a one-in-three risk for either. P(NPV > 0) is **99.8 and 100 %** for steam at CS1 and CS3 (P10 NPV +AUD 3.8 M and +5.2 M), 89.7 and 91.3 % for tCO_2, and 72.2 and 67.7 % for steam and tCO_2 at CS2. The sCO_2 cycle is unviable in practice: 0.5 % at CS2, 33 % at CS1 and 49 % at CS3, with negative median NPV at all three.

Finding 27 — Cheaper heat exchangers would make tCO_2 competitive with steam only at the 200 °C site, and would not rescue sCO_2 anywhere. A **11 % reduction** in the cost of the tCO_2 heater and recuperator

would bring its LCOE level with steam at CS2, where its recovery advantage is largest; at CS1 a 95 % reduction would be needed, and at CS3 even free heat exchangers would leave it above steam, because the remaining turbomachinery and heat-rejection costs are themselves above the whole steam plant. For sCO_2 the target is not reachable at any site. The competitive window for CO_2 cycles in Australian WHR is therefore the low-temperature, deep-recovery case, and it depends on a cold heat sink (Finding 12).

6.4 Summary of RQ3 Results

Sized to what each cycle can actually recover from the site stream and costed on a consistent basis, the steam Rankine cycle gives the lowest cost of electricity at all three sites and the highest NPV at the two sites above 300 °C, with discounted paybacks of 3–4 years and IRRs of 28–36 %. The transcritical CO_2 cycle is the better investment at the 200 °C Pilbara site, where its deep heat recovery earns it 53 % more NPV than steam, but only if a 14 °C heat sink can be provided at no cost, which Finding 12 shows it cannot. The supercritical CO_2 cycle is not viable at any of the three sites under 2026 conditions. The central lesson is that the right figure of merit for a WHR cycle is the work recovered per kilogram of waste gas, and its cost; thermal efficiency alone predicts neither.

7. Off-Design and Part-Load Performance (RQ4)

This chapter answers RQ4. Section 7.1 reports efficiency retention at five load levels for each cycle at each site design point, and Section 7.2 the Marchionni flexibility scores.

7.1 Part-Load Efficiency Curves

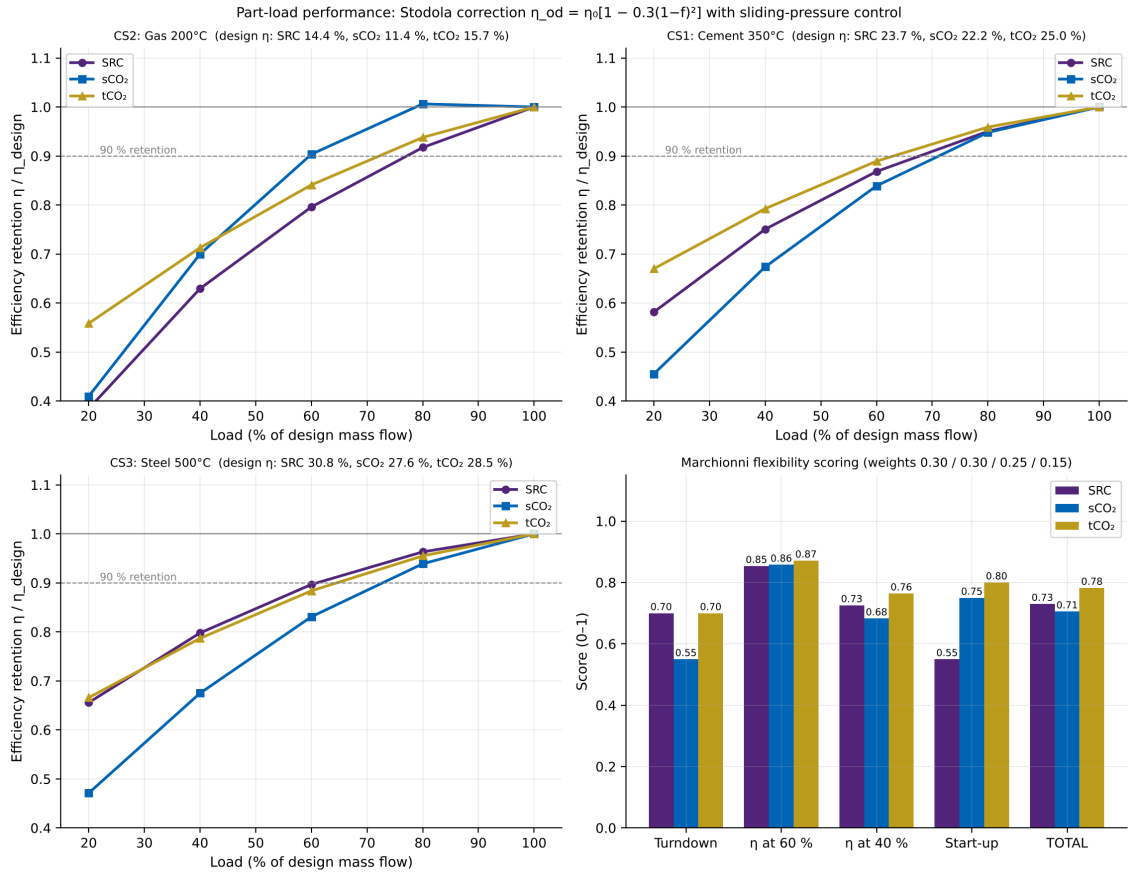


Figure 7.1: Efficiency retention versus load at the three site design points (Stodola correction with sliding-pressure control) and the Marchionni flexibility scores.

Table 7.1: Thermal efficiency (%) at part load, and retention relative to the design point (in parentheses).

Site	Cycle	100 %	80 %	60 %	40 %	20 %
CS2	SRC	14.44	13.24 (0.917)	11.49 (0.796)	9.09 (0.629)	5.55 (0.384)
CS2	sCO_2	11.43	11.50 (1.006)	10.33 (0.904)	8.00 (0.700)	4.68 (0.410)
CS2	tCO_2	15.67	14.70 (0.938)	13.18 (0.841)	11.17 (0.713)	8.75 (0.558)
CS1	SRC	23.67	22.49 (0.950)	20.55 (0.868)	17.76 (0.751)	13.76 (0.582)
CS1	sCO_2	22.24	21.09 (0.948)	18.66 (0.839)	14.99 (0.674)	10.13 (0.455)
CS1	tCO_2	24.97	23.95 (0.959)	22.21 (0.889)	19.79 (0.792)	16.73 (0.670)
CS3	SRC	30.82	29.68 (0.963)	27.63 (0.897)	24.59 (0.798)	20.21 (0.656)
CS3	sCO_2	27.58	25.89 (0.939)	22.91 (0.831)	18.61 (0.675)	12.99 (0.471)
CS3	tCO_2	28.49	27.21 (0.955)	25.18 (0.884)	22.42 (0.787)	18.97 (0.666)

Finding 28 — The tCO_2 cycle retains the most efficiency at deep part load; the sCO_2 cycle loses the most. At 20 % load tCO_2 retains **56–67 %** of its design efficiency, steam 38–66 % and sCO_2 41–47 %. The tCO_2 advantage comes from the liquid pump, whose work falls in proportion to flow while the Stodola penalty applies only to the turbine, and from the gentler sliding-pressure schedule $(0.6 + 0.4f)$ that a pumped cycle can tolerate. The sCO_2 cycle carries the Stodola penalty on both its turbine and its compressor, and because the compressor already absorbs 31–64 % of turbine work at design, a proportionate loss in both machines erodes the net output fastest.

Finding 29 — The steam cycle's part-load behaviour depends strongly on its design pressure: it is the most robust cycle at 500 °C and the least at 200 °C. At CS3, where the boiler runs at 3.7 MPa, steam retains 90 % at 60 % load and 66 % at 20 % load, marginally better than tCO_2. At CS2 the pinch-limited boiler pressure is only 0.16 MPa, and sliding it to 0.03 MPa at 20 % load leaves little expansion ratio above the 10 kPa condenser: retention falls to **38 %**. Low-grade steam WHR is therefore poorly suited to a fluctuating source.

Finding 30 — The apparent efficiency *gain* of sCO_2 at 80 % load at the 200 °C site is a pressure-ratio effect, not a near-critical effect. At CS2 the sCO_2 retention at 80 % load is 1.006. The sliding-pressure schedule

lowers the high-side pressure from 20 to 18 MPa, and at a 190 °C turbine inlet the design pressure ratio of 2.5 is above the efficiency-optimal ratio, so the reduction helps more than the Stodola penalty hurts. The compressor inlet state is fixed at 36 °C and 8 MPa at all loads, so the Interim Report's attribution of this feature to near-critical compressor behaviour is withdrawn. The effect vanishes at 350 and 500 °C, where the design pressure ratio is closer to optimal.

7.2 Operational Flexibility Scoring

Table 7.2: Marchionni flexibility scores (retention values averaged over the three sites).

Cycle	Turndown (0.30)	η at 60 % (0.30)	η at 40 % (0.25)	Start-up (0.15)	Total
SRC	0.70	0.854	0.726	0.55	0.730
sCO_2	0.55	0.858	0.683	0.75	0.706
tCO_2	0.70	0.871	0.764	0.80	**0.782**

The tCO_2 cycle scores highest (0.782), steam second (0.730) and sCO_2 third (0.706). The ordering is the same as the Interim Report's, but the margins are smaller (the interim scores were 0.854, 0.763 and 0.801 for tCO_2, steam and sCO_2) because the corrected recuperator reduces the part-load advantage of both CO_2 cycles and the pinch-limited steam pressures change the steam curves. The tCO_2 lead rests on three properties: a liquid pump, the simplest start-up of the three (no steam drum, no near-critical compressor control), and the best efficiency retention at 60 and 40 % load.

7.3 Summary of RQ4 Results

For a fluctuating heat source the transcritical CO_2 cycle is the most flexible of the three, retaining 56–67 % of its design efficiency at one-fifth load and scoring 0.78 on the Marchionni scale against 0.73 for steam and 0.71 for sCO_2. The steam cycle is nearly as flexible at high source temperature but degrades badly at 200 °C, where its boiler pressure is already very low. The sCO_2 cycle degrades fastest at deep part load at every site. Flexibility is the one criterion on which tCO_2 is unambiguously preferred, and it is most valuable precisely at the batch-type sources (electric arc

furnaces, intermittent process trains) where it matters.

8. Discussion

The four results chapters tell one story. Under consistent, second-law- consistent assumptions and with the waste-gas stream treated as what it is — a fixed resource at a warm site — the steam Rankine cycle is the most efficient, the cheapest and the most bankable option at 350 and 500 °C, the transcritical CO_2 cycle recovers the most energy and is the most flexible option but needs a cold heat sink that Australia does not naturally provide, and the supercritical CO_2 Brayton cycle, in its simple recuperated form, is not competitive below 500 °C at all. This chapter explains why these conclusions differ from much of the published literature and from this project's own Interim Report, sets out the figure of merit that should be used for WHR cycle selection, develops the Australian heat-sink constraint, and states the limitations within which the conclusions hold.

8.1 Why the Literature's CO_2 Advantage Did Not Appear

Four boundary conditions account for the difference between this study and the 5–15 point efficiency advantages reported for CO_2 cycles by Kizilkan [13], Jiang et al. [11] and Crespi et al. [9]; each is quantified in Chapter 4.

1. **Recuperator energy balance (Finding 3).** With the effectiveness applied to the limiting stream, recuperation adds 3–10 points to the tCO_2 cycle, not 12–20. Near the critical point the high-pressure stream can absorb two to three times more enthalpy per kelvin than the low-pressure stream can supply; a cold-side-only effectiveness transfers heat the exhaust does not have. The Interim Report made this error, and some published models that report unusually high recuperated efficiencies may as well.

2. **Heat sink temperature (Findings 11–13).** The sCO_2 cycle's efficiency rests on compressing a dense near-critical fluid; at a 36 °C compressor inlet, which is what Australian cooling water allows, the fluid is already gas-like and the back-work ratio is 31–64 % at the three sites. Moving the inlet from 32 to 40 °C removes two-fifths of the cycle's efficiency at 200 °C. The tCO_2 baseline condenses at 14 °C, which no Australian site can supply without refrigeration.

3. **Steam baseline.** Many comparisons evaluate steam at a single modest pressure. At turbine inlet temperatures of 350–500 °C a 10 MPa steam cycle with

a 10 kPa condenser reaches 32–34 % and is a demanding baseline (Finding 1). When the boiler is integrated with the stream the steam pressure falls to 0.9–3.7 MPa and the site efficiency to 24–31 % (Finding 14), close to Kizilkan's 24 %; the CO_2 cycles then compete on recovery depth, not on efficiency.

4. **Cycle layout.** The simple recuperated sCO_2 layout modelled here is the standard single-loop configuration. Its recuperated heater inlet of 100–279 °C means it recovers only 60–65 % of the heat that the tCO_2 cycle recovers from the same stream (Finding 15). Cascade, dual-recuperated and preheating layouts that address this were outside the scope of a consistent three-cycle comparison and are the obvious next step (Section 9.3).

The disagreement, then, is not with the thermodynamics of CO_2 but with the conditions under which it is attractive: a cold sink, a high source temperature, and a layout designed for deep recovery. Australian industrial WHR at 200–500 °C offers none of the first two.

8.2 The Figure of Merit for Waste Heat Recovery

Thermal efficiency is the wrong single number for WHR cycle selection. The owner of a 50 kg/s cement exhaust does not care what fraction of the heat entering the cycle becomes electricity; they care how much electricity the exhaust yields and what the plant costs. Equation (3.15), $w_{gas} = \eta_{th}\, c_{p,gas}(T_{src} - T_{gas,out})$, captures the first quantity, and the results of Chapters 5 and 6 show that the second factor can outweigh the first: tCO_2 at CS2 is 1.2 points more efficient than steam but recovers 75 % more power; sCO_2 at CS1 is 1.4 points less efficient than steam but recovers 25 % less.

The second quantity, cost, must then be evaluated at the output the cycle actually delivers, because specific capital cost falls steeply with size (Table 5.3). A fixed-output comparison at 1 MWe — the basis of the Interim Report and of much of the literature — reduces the whole economic analysis to the capital cost ranking, because with free heat and a common life and capacity factor LCOE is proportional to TEC/kW (Finding 25). The fixed-source framing used here is the one a feasibility study would use, and it is the one on which tCO_2 earns its single economic win, at the 200 °C site.

8.3 Heat Rejection in a Warm Climate

The decisive Australian constraint is the heat sink. Three results combine:

- The tCO_2 cycle at $P_{low} = 5\,\text{MPa}$ condenses at 14.3 °C and needs a sink near 4 °C (Finding 11). Cooling-tower water at Gladstone, Sydney or the Pilbara is 20–32 °C for most of the year [33]; seawater at the Pilbara coast exceeds 25 °C.

- Supplying that sink with an electric chiller would consume 63–134 % of the cycle's net output at COP 4 (Finding 12). The tCO_2 NPV advantage at CS2 (Finding 23) therefore cannot be realised there; it exists only where a natural cold sink is available — cold river or groundwater, a deep-water intake, or a co-located refrigeration load.

- Raising P_{low} to 7 MPa (condensing at 28.7 °C) makes the cycle compatible with evaporative cooling outside summer but costs 23–30 % of the recovered work; at CS1 the result is exactly the steam cycle's 52.8 kJ per kilogram of gas, at a higher capital cost. At the Pilbara, with summer wet-bulb temperatures above 28 °C, even 7 MPa is marginal.

The exergy analysis flags the same issue from the other direction: the negative condenser term in Table 4.3 is the unpaid refrigeration exergy. The sCO_2 cycle faces a milder version of the constraint at its compressor inlet (Finding 13). The steam cycle, condensing at 45.8 °C, is the only one of the three that is indifferent to Australian ambient conditions, and this, more than any thermodynamic property, is why it emerges as the default choice.

8.4 Cycle Selection Framework for Australian Industry

Table 8.1 condenses the thirty findings into a selection guide for a 0.3–3.5 MWe plant at 2026 costs and prices.

Table 8.1: Recommended cycle selection for Australian industrial WHR (2026 conditions, fixed-source basis).

Source	Recommend	Basis and conditions
$\geq 450\,^{\circ}\mathrm{C}$ (EAF, kiln)	SRC	LCOE AUD 31/MWh, NPV 12.4 M, payback 3.2 yr, 100 % Monte Carlo viability; steam is also the most efficient cycle here. tCO_2 only if load is highly variable and a cold sink exists.
300–450 °C (cement)	SRC	LCOE AUD 35/MWh, NPV 11.4 M, payback 4.2 yr, 99.8 % viability. tCO_2 recovers 29 % more power but costs 94 % more; its NPV is 12 % lower and its sink requirement is unmet.
150–300 °C (gas processing)	SRC, or tCO_2 with a natural cold sink	Both marginal: LCOE 96–100 against a 130 AUD/MWh price, payback 10–11 yr, 68–72 % viability. tCO_2 gives 53 % more NPV where water below $\approx$10 °C is available; otherwise steam, or no project.
Any temperature	Not sCO_2 (simple recuperated)	Negative or marginal NPV at every site; $P(\mathrm{NPV} > 0)$ 0.5–49 %. Reconsider only with a cascade layout, a compressor inlet below 33 °C, and sources above 500 °C.
Fluctuating load	tCO_2 preferred	Marchionni score 0.78 against 0.73 (SRC) and 0.71 (sCO_2); steam is weakest at 200 °C.

Three qualifications apply. First, the steam recommendation at 200 °C is weak: with an IRR of 12 % against a 7 % discount rate and a 28 % chance of loss, the project depends on the Pilbara price premium and would not proceed at NEM prices. Second, the tCO_2 option at any site should be evaluated with the actual sink: with cold water the cycle is the best choice at 200 °C; with a chiller it is not viable; at 7 MPa it is a more expensive equivalent of steam. Third, all three cycles are small plants in cost terms — heat exchangers are 35–43 % of CO_2 equipment cost and heat rejection 17–44 % of every cycle's — so the recommendation is sensitive to the heat exchanger correlations, which is why the Monte Carlo range of ± 30 % on capital was carried through.

8.5 Limitations and Assumptions

L1 — Constant turbomachinery efficiencies. $\eta_t = 0.85$ and $\eta_{p/c} = 0.80$ are applied at all sizes and all three cycles. Real sub-MW machines are less efficient, and the penalty is probably larger for the high-pressure CO_2 turbines than for steam.

L2 — No pressure drop. All exchangers are isobaric. PCHE pressure drops of 1–2 % of the high-side pressure would reduce the CO_2 cycle efficiencies by roughly one point and raise the steam cycle's by nothing.

L3 — Cost correlations extrapolated. The Weiland turbine and generator correlations begin at 8 and 4 MW; the Turton compressor at 450 kW. Their stated uncertainties (−32 to +51 % on the turbine) are wider than the ±30 % carried in the Monte Carlo. The steam turbine benchmark is for back-pressure units; a condensing turbine with VFR of 45–148 would cost more, which would narrow but not close the gap to tCO_2 at CS1 and CS3.

L4 — Heat rejection modelled uniformly. A 10 K mean temperature difference and the Weiland air-cooler correlation are applied to all three cycles. This is conservative for the 46 °C steam condenser (cheaper water-cooled units exist) and optimistic for the 14 °C tCO_2 condenser (no chiller is costed; Finding 12 quantifies the omission).

L5 — CO_2 pressures not re-optimised for recovery. The CO_2 high- and low-side pressures follow the RQ1 rules, whereas the steam pressure was optimised for w_{gas} at each site. A tCO_2 high-pressure optimisation on the same basis would improve its results by a few per cent.

L6 — Simple cycle layouts. One layout per cycle, no reheat, recompression, cascade or split flow. This is the right basis for a like-for-like comparison but understates what an optimised sCO_2 system could do above 500 °C.

L7 — Monte Carlo scope. Capital, price and capacity factor are varied; source temperature and flow, discount rate, O&M and plant life are not. Correlation between variables is ignored.

L8 — Part-load model. A single parabolic Stodola coefficient calibrated to one tCO_2 dataset [27] is applied to all three cycles, and the heater is not re-integrated at part load.

L9 — Economic scope. No carbon price, no grid-connection or site-integration costs, no availability penalty for the immature CO_2 technologies, and an 85 % capacity factor that may be optimistic for batch processes.

None of these limitations favours the CO_2 cycles systematically except L4 (no chiller) and L3 (back-pressure steam benchmark), which act in opposite directions; L1, L2 and L3 (turbine extrapolation) act against them. The qualitative conclusions — steam first above 300 °C, tCO_2 viable only with a cold sink, simple sCO_2 not viable — are unlikely to change within these uncertainties; the margins would.

8.6 Comparison With Published Literature

The present results agree with Baglietto et al. [14] that the mature steam supply chain gives steam a capital cost advantage that CO_2 cycles do not overcome at industrial WHR scale, and extend that finding to the smaller 0.3–3.5 MWe range and to Australian prices. They agree with Kizilkan [13] on the steam efficiency at a 350 °C cement source once the boiler is integrated with the stream, but not on the sCO_2 advantage, for the reasons given in Section 8.1. They agree in direction with Jiang et al. [11] that tCO_2 outperforms sCO_2 at low temperature and with Yang et al. [27] and Marchionni et al. [17] that pumped CO_2 cycles have the better part-load behaviour. The observation that recuperated sCO_2 cycles recover a waste-heat stream shallowly is well known in the sCO_2 community and is the reason for the cascade and split-flow WHR layouts described by Crespi et al. [9]; this thesis quantifies the penalty for the simple layout against two alternatives on one basis. The heater-integration result (Findings 6 and 18) is, to the author's knowledge, the first explicit demonstration that the pseudo-critical c_p peak produces a double-pinched, mid-bulging profile rather than an interior pinch, and that two-point LMTD sizing overestimates the tCO_2 heater by a factor of two to three at $\varepsilon = 0$.

8.7 Revisions Since the Interim Report

For transparency, the five corrections made between the Interim Report and this thesis, and their consequences, are listed in Table 8.2. Each was found by re-checking the model code against its own outputs after the Interim Report was marked.

Table 8.2: Corrections made after the Interim Report.

Correction	What was wrong	Consequence
Recuperator energy balance (Eq. 3.5)	Effectiveness applied to the cold stream only; hot stream cooled below the cold inlet (second-law violation) at every temperature for both CO_2 cycles.	CO_2 efficiencies fall 3–11 points; sCO_2/SRC crossover at 362 °C and "tCO_2 beats SRC everywhere" withdrawn (Finding 1).
Heater gas profile (Eq. 3.12)	Gas temperature linear in node index, not in heat duty, in the pinch diagrams; gave a constant 10 K gap and a spurious "mid-exchanger pinch at 129 °C".	Double pinch at both terminals with a mid-exchanger bulge (Finding 6). LMTD comparison (Finding 18) unaffected, as it already used the correct profile.
Heater sizing in the cost model	sCO_2 heater driving force computed from an expression that cancelled to −10 K (giving 3,400–7,000 K); steam boiler driving force from an ad hoc terminal estimate; tCO_2 heater at the worst-case 10 K.	sCO_2 no longer the cheapest cycle; steam cheapest at every site (Finding 19).
Cost basis	Turbine coefficients not traceable to the cited source; condenser omitted; compressor costed as a pump.	Consistent correlations (Table 3.2); heat rejection 17–44 % of equipment cost (Finding 21).
Economic basis	Fixed 1 MWe output regardless of stream; simple payback; steam at $0.95\,P_{sat}$ at a 200 °C source (saturation 2 K below the source).	Fixed-source basis with pinch-limited steam pressure; discounted payback; tCO_2 NPV lead at 200 °C (Finding 23).

9. Conclusions and Recommendations

9.1 Answers to the Research Questions

9.1.1 RQ1: Thermodynamic Performance

Under consistent assumptions — equal turbomachinery efficiencies, a recuperator limited by both streams, and a 36 °C sCO_2 compressor inlet consistent with Australian cooling water — the steam Rankine cycle has the highest thermal efficiency at every turbine inlet temperature from 150 to 600 °C (19.1–35.4 %), followed by tCO_2 (13.3–30.1 %) and sCO_2 (5.3–30.1 %); the sCO_2 cycle is unviable below 300 °C because its compressor absorbs 45–80 % of turbine work. Per unit of net work steam is also the most exergy-efficient cycle at a common turbine inlet temperature; at realistic pinch-limited boiler pressures tCO_2 leads at 200 and 350 °C. The tCO_2 heater is pinched at both terminals with the driving force widest in the middle, so the pseudo-critical c_p peak enlarges the mean temperature difference rather than restricting it. Recuperation raises efficiency by 3–10 points at $\varepsilon = 0.70$ but raises the gas outlet temperature with it, so it adds only 8–18 % to the work recovered per kilogram of gas. The tCO_2 baseline needs a 14 °C condenser that a chiller could supply only by consuming 63–134 % of the net output, and the sCO_2 cycle loses two-fifths of its 200 °C efficiency if its compressor inlet warms from 32 to 40 °C.

9.1.2 RQ2: Equipment Sizing and Capital Cost

CO_2 turbines are 15–26 times more compact than the steam turbine at the same site and remain single-stage (VFR 2.0–3.1) where steam needs multi-stage machines above 350 °C; this advantage is robust. Integrated with the site stream, the steam boiler must run at 0.16, 0.91 and 3.69 MPa, and the tCO_2 cycle recovers the most power at 200 and 350 °C (723 and 3418 kW against 412 and 2639 kW for steam) because its cold heater inlet cools the gas furthest; sCO_2 recovers the least at every site. At equal output steam is the cheapest plant at every site (AUD 4,444, 3,071 and 2,598/kW at 1 MWe), tCO_2 costs 30–65 % more and sCO_2 two to three times more, the latter because its back-work ratio requires 2.8 MW of turbine and 1.8 MW of compressor per megawatt delivered at 200 °C. Two-point LMTD sizing overestimates the tCO_2 heater area by 129–223 % when the c_p peak lies inside it.

9.1.3 RQ3: Australian Techno-Economic Analysis

On a fixed-source basis the steam cycle gives the lowest LCOE at all three sites (AUD 96, 35 and 31/MWh) and the highest NPV at 350 and 500 °C (AUD 11.4 M and 12.4 M, discounted payback 4.2 and 3.2 years, IRR 28 and 36 %, 99.8–100 % Monte Carlo viability). The tCO_2 cycle is the better investment at 200 °C (NPV 1.71 M against 1.12 M) because it recovers 75 % more power, but only where a natural cold sink exists; both options there are marginal (payback 10–11 years, 68–72 % viability). The sCO_2 cycle is not viable at any site (NPV negative at CS2 and CS1; 0.5–49 % viability). Cheaper heat exchangers would bring tCO_2 level with steam at 200 °C with an 11 % reduction, but not at 350 or 500 °C, and would not rescue sCO_2 anywhere.

9.1.4 RQ4: Off-Design Performance

The tCO_2 cycle is the most flexible (Marchionni score 0.78; 56–67 % efficiency retention at 20 % load), steam second (0.73; 38–66 %, worst at the low-pressure 200 °C design) and sCO_2 third (0.71; 41–47 %). The sCO_2 efficiency rise at 80 % load at 200 °C is a pressure-ratio effect of sliding-pressure control, not a near-critical effect.

9.2 Original Contributions

1. **A consistent three-cycle Australian benchmark.** All three cycles modelled with the same property library, turbomachinery efficiencies, recuperator definition, heater integration, cost correlations and economic basis, at three real Australian sites, with every number regenerated from one script. No published study does this for SRC, sCO_2 and tCO_2 together.

2. **Quantification of why the CO_2 advantage is conditional.** The thesis isolates four boundary conditions — recuperator energy balance, heat-sink temperature, steam baseline and cycle layout — and quantifies each: a cold-side-only recuperator overstates tCO_2 efficiency by 9–10 points; a 32→40 °C compressor inlet removes 42 % of sCO_2 efficiency at 200 °C; a 14 °C condenser costs more to provide than the tCO_2 cycle produces at 200 °C.

3. **Corrected heater integration for near-critical CO_2.** With a duty-linear gas profile the tCO_2 heater exhibits a double pinch with a mid-exchanger bulge of 42–65 K, and two-point LMTD overestimates its area by 129–223 % at $\varepsilon = 0$. The thesis also shows that the heater inlet temperature, raised by recuperation, is the floor on heat recovery.

4. **A fixed-source figure of merit.** Net work per kilogram of waste gas (Eq. 3.15), combined with cost at the delivered output, is shown to rank cycles differently from thermal efficiency, and to be the quantity on which the only CO_2 economic win (tCO_2 at 200 °C) rests.

5. **A selection framework for Australian industry** validated against 10,000-sample Monte Carlo uncertainty and transparent about its revisions from the interim stage.

9.3 Recommendations for Future Work

1. **Cascade and split-flow sCO_2 layouts** for deep recovery, evaluated on the same fixed-source basis, since the simple recuperated layout's shallow recovery is its principal economic handicap.

2. **Joint optimisation of CO_2 pressures and recuperator effectiveness for** w_{gas}, as was done for the steam pressure here, with the chiller or cooling-tower system costed explicitly as a function of site wet-bulb temperature.

3. **Vendor-based costs for sub-5 MW CO_2 turbomachinery and PCHEs**, including pressure drop, to replace the extrapolated Weiland correlations; and a condensing-turbine cost for steam to replace the back-pressure benchmark.

4. **Dynamic simulation** with turbomachinery maps and heater re-integration at part load, to replace the single-coefficient Stodola model.

5. **Identification of Australian sites with natural cold sinks** (cold groundwater, deep-water intakes, LNG regasification cold) where the tCO_2 cycle's recovery advantage could be realised.

6. **Lifecycle greenhouse-gas and carbon-price sensitivity**, which would favour the cycle that recovers the most energy rather than the one with the lowest capital cost.

9.4 Concluding Statement

Industrial waste heat recovery in Australia is a sound investment above 300 °C: at 2026 costs and prices a steam Rankine plant at a cement preheater or an electric arc furnace returns its capital in three to four years and is profitable in essentially every scenario examined. The CO_2 cycles do not displace it. Their compactness is

real and their part-load behaviour is better, but at Australian sink temperatures the supercritical Brayton cycle loses the near-critical compression on which its efficiency depends, and the transcritical Rankine cycle — which recovers more energy than steam from any stream below 400 °C — needs a condenser colder than the country can supply without spending more on refrigeration than the cycle earns. The thermodynamics that make CO_2 attractive in the literature are not wrong; they belong to cold climates, hot sources and optimised layouts. For an Australian operator in 2026 the recommendation is steam at 350 °C and above, a careful site- specific study at 200 °C, and transcritical CO_2 wherever a natural cold sink makes its deep recovery usable.

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A. Python Model Code

All models are contained in the Python package `co2whr` and four run scripts, supplied with this thesis. Running `python run_all.py` from the `code/` directory regenerates every figure in `figures/` and writes every number used in this thesis to `results/results.json` and `results/summary.txt` (about two minutes on a laptop). Python 3.9+, CoolProp 6.x (HEOS backend), NumPy and Matplotlib are required.

`co2whr/cycles.py` Steady-state state-point models for the SRC, sCO_2 Brayton and tCO_2 Rankine cycles with a recuperator limited by both streams (Eq. 3.5); flow exergy and component exergy balance.

`co2whr/heatex.py` Counter-flow heater integration with a duty-linear gas profile, bisection for the gas outlet temperature at a 10 K approach, segmented UA (Eq. 3.14) and two-point LMTD; recuperator UA with real properties on both sides; cooler UA.

`co2whr/design.py` Site design points: pinch-limited steam pressure by golden-section search on w_{gas}, recovered power from each stream, LMTD-versus-segmented comparison.

`co2whr/costs.py` Cost correlations of Table 3.2, CEPCI escalation, currency conversion and total erected cost by component.

`co2whr/economics.py` LCOE, NPV, discounted and simple payback, IRR, Monte Carlo sampling and heat-exchanger breakeven.

`co2whr/offdesign.py` Stodola part-load correction with sliding-pressure control; Marchionni flexibility scoring.

`co2whr/sites.py` The three case-study definitions and shared constants.

`run_rq1.py` – `run_rq4.py`, `run_all.py` One script per research question producing the figures and result tables; `run_all.py` runs all four and writes the results files.

Validation outcomes: SRC model versus NIST Webbook, maximum deviation 0.031 % on specific enthalpy; CO_2 critical point versus NIST REFPROP, 0.04 %; global exergy balance closure below 10^{-12} kJ/kg for all three cycles; recuperator internal tem-

perature difference positive at every node for every reported case. The Interim Report model files (`tco2_model.py`, `sco2_model.py`, `src_model.py`, `rq2_sizing.py`, `rq2_cost.py`, `rq3_economics.py`, `rq4_offdesign.py`) are retained in the repository under `code/interim_original/` for comparison; their known errors are listed in Table 8.2.

B. Complete Findings Log

All thirty findings are listed below with their section reference and key numerical value.

No.	Finding	Key value
RQ1 — Thermodynamic performance (Chapter 4)		
F1	SRC most efficient at every TIT, 150–600 °C	SRC 19.1–35.4 %; tCO_2 13.3–30.1 %; sCO_2 5.3–30.1 %; no crossover
F2	sCO_2 BWR prohibitive below 300 °C	79.8 % at 150 °C; 45 % at 300 °C; 26.9 % at 600 °C
F3	Recuperator gain bounded by the hot stream	+3.4/+7.0/+10.1 pp at $\varepsilon = 0.70$ (interim: +12.3/+16.8/+20.3)
F4	Steam boiler largest single loss; steam still most exergy-efficient per unit work at 400 °C	375 kJ/kg (56.6 %); 0.70 vs 1.09 (tCO_2) vs 1.25 (sCO_2) kJ/kJ
F5	Recuperator the dominant CO_2 irreversibility	28 % (sCO_2) and 32 % (tCO_2) of destruction at 400 °C
F6	tCO_2 heater pinched at both terminals, bulge mid-exchanger	ΔT_{max} 41.6/49.0/64.7 K at 88/145/166 °C
F7	Heater inlet sets the floor on recovery; recuperation raises it	Gas out 34–42 °C ($\varepsilon = 0$) vs 60/101/151 °C ($\varepsilon = 0.70$)
F8	Pseudo-critical peak sharpens as pressure falls	4.42 kJ/kg·K at 57 °C (12.8 MPa); 2.62 at 76 °C (20 MPa)
F9	Recuperation adds more to efficiency than to recovered work	ε 0→0.85: η +4.5/+9.3/+13.9 pp; w_{gas} +12/+16/+21 %
F10	High ε: recuperator approaches heater duty and the approach limit	302 vs 399 kJ/kg at CS3; ΔT_{min} 9.8 K at CS2
F11	Low condensing pressures climatically infeasible	3.5/4.0/5.0 MPa condense at 0.2/5.3/14.3 °C

No.	Finding	Key value
F12	Chiller for a 14 °C sink would consume most or all of the output	134/75/63 % of net work at COP 4
F13	Warm sink penalty on sCO_2 concentrated at low TIT	32→40 °C inlet: 15.7→9.2 % at 200 °C
RQ2 — Sizing and capital cost (Chapter 5)		
F14	Integrated steam pressure far below the sweep value	0.16/0.91/3.69 MPa; site η 14.4/23.7/30.8 %
F15	tCO_2 recovers the most power at 200 and 350 °C; sCO_2 the least everywhere	723/3418/2186 kW vs SRC 412/2639/2273 vs sCO_2 340/1988/1280
F16	CO_2 turbines 15–26× more compact	SP 0.63–0.81 (SRC) vs 0.031–0.042 (CO_2)
F17	Steam needs multi-stage turbines above 350 °C; CO_2 single-stage	VFR 10.5/45/148 (SRC) vs 2.0–3.1 (CO_2)
F18	Two-point LMTD overestimates tCO_2 heater area	+129/+171/+223 % at $\varepsilon = 0$
F19	Steam cheapest per kW at every site	4,444/3,071/2,598 AUD/kW at 1 MWe; tCO_2 +30–65 %; sCO_2 2.3–2.6×
F20	sCO_2 penalty is turbomachinery	2.8 MW turbine + 1.8 MW compressor per MWe at CS2; 37 % of equipment
F21	Heat exchangers 35–43 % of CO_2 equipment; heat rejection 17–44 % for all	Cooler largest for 14 %-efficient steam at CS2
RQ3 — Techno-economics (Chapter 6)		
F22	Steam lowest LCOE at every site	96/35/31 AUD/MWh; tCO_2 100/53/57; sCO_2 243/94/91
F23	tCO_2 best NPV at 200 °C; steam at 350 and 500 °C	CS2: 1.71 vs 1.12 M; CS1: 11.4 vs 10.1 M; CS3: 12.4 vs 7.5 M
F24	Discounted payback 3–4 yr steam, 7 yr tCO_2 at hot sites; 10–11 yr at CS2	IRR 28/36 % (SRC), 17/18 % (tCO_2), 12 % at CS2
F25	Economics follow efficiency × recovery depth	sCO_2 at CS1: −1.4 pp but −25 % power; tCO_2 at CS2: +1.2 pp but +75 % power

No.	Finding	Key value
F26	Steam robust at 350/500 °C; CS2 a one-in-three risk	$P(\mathrm{NPV} > 0)$ 99.8/100 % (SRC), 90/91 % (tCO_2), 72/68 % at CS2
F27	Cheaper HX rescues tCO_2 only at 200 °C	11 % cut at CS2; 95 % at CS1; unreachable at CS3 and for sCO_2
RQ4 — Off-design (Chapter 7)		
F28	tCO_2 retains most efficiency at deep part load; sCO_2 least	56–67 % vs 38–66 % (SRC) vs 41–47 % (sCO_2) at 20 % load
F29	Steam part-load behaviour depends on design pressure	66 % retention at CS3 (3.7 MPa) vs 38 % at CS2 (0.16 MPa)
F30	sCO_2 80 %-load gain at CS2 is a pressure-ratio effect	Retention 1.006; disappears at 350/500 °C

C. AI Assistance Log

This log records the use of generative AI tools in this thesis, as required by UQ policy [36]. Each entry states the section affected, the tool, what was requested, what was received and how it was used.

Section	Tool	Request	Output and use
Ch. 1–2	Claude	Draft background and literature-review paragraphs from my reading notes and reference list	Draft prose; checked against the primary sources, reorganised and rewritten where needed
Ch. 1–3	Claude	LaTeX formatting of definitions, equations, tables and the methodology flow diagram	Typesetting only; content from my proposal and interim report
Model code (interim)	Copilot	Autocomplete while writing `tco2_model.py` and the other interim scripts	CoolProp/NumPy/Matplotlib syntax completions; all model logic written by me
Model code (final)	Claude	Review my seven interim Python files and their printed outputs	Identified: recuperator limited by the cold stream only (second-law violation); gas profile linear in node index in the pinch diagrams; cancelling temperature-difference expression for the sCO_2 heater and ad hoc boiler ΔT in `rq2_cost.py`; condenser cost defined but unused; turbine coefficients not traceable to the cited paper; simple payback; Monte Carlo not varying source temperature as claimed

Section	Tool	Request	Output and use
Model code (final)	Claude	Restructure the scripts into one package implementing the corrections, a consistent heater integration, the cost basis of Table 3.2, fixed-source economics, discounted payback and breakeven	`co2whr` package and run scripts (Appendix A). I ran the package, checked balance closures, verified every correlation against Weiland et al. (2019), Turton et al. (2012) and DOE (2012), and compared outputs with my interim results
Sec. 3.6.4	Claude	Locate and verify the Weiland et al. (2019) correlation coefficients	Coefficients and validity ranges from the NETL/DOE source; used in Table 3.2 after my own check of the paper
Figures	Claude	Matplotlib scripts for all figures from the model outputs	Figures regenerated by `run_all.py`; captions written with the chapter text
Ch. 4–9	Claude	Draft the results, discussion and conclusion chapters from `results/summary.txt` and my notes on interpretation	Draft text with finding boxes; I checked every number against the results file, revised the interpretation and approved the final wording
Abstract, Exec. Summary, Declaration	Claude	Draft from the final chapters	Revised and approved by me
Conference slides	Claude	Build and revise the presentation from the thesis content and supervisor feedback	Slides and speaker notes; presented by me on 8 October 2026